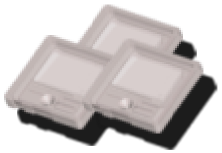
UriSed 3 PRO

Service training

Budapest, 27 September 2018
E77 Service Team

Introducing the UriSed 3 PRO



Cuvette and UriSed 3 PRO Parameters

UriSed 3 PRO cuvettes:

- **Package: 50 pcs/container, 12 container/box**
- **Max. cuvette load in sediment analyzer: 600 pcs**
- **Sample volume: 200 µl**

Detected particles:

- **RBC /RBC-ACA, RBC-g**
- **WBC**
- **Crystals /CaOxd, CaOxm, Triple phosphate, Uric acid**
- **Hyaline casts (HYA)**
- **Pathological casts (PAT)**
- **Non-squam. Epith. Cell (NEC)**
- **Squamous Epith. Cell (EPI)**
- **Yeast (YEA)**
- **Bacteria (BAC)**
- **Mucus MUC)**
- **Amorphous Material (AMO)**

System installation

Hardware installation
1. Unpacking
2. Installation of Main Unit
   ->Remove reatiner bolt
   ->Rotor placement
   ->Front Cuvette Rail placement
   ->Back Cuvette Rail placement
3. Fluidic system
   ->Water container and connections
   ->Waste container and connections
4. Rack conveyor
   ->Installation
   ->Level adjustment
5. PC and accessories
6. Switch Device On

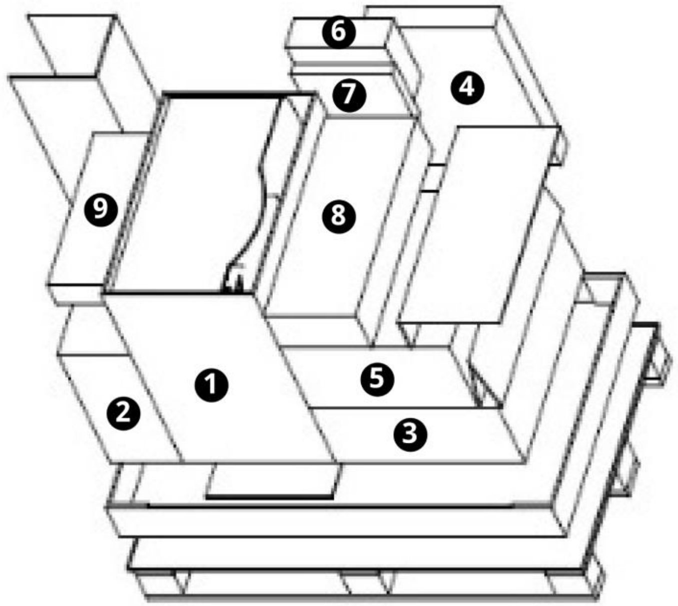
Instrument Setup
1. Disable the interlock
2. Start the Service software (systest)
3. Rack puller adjustment
4. Cuvette fillup position setup
5. Urine removing setup
6. Stub position setup
7. Cuvette detection laser position setup
8. Centrifuge null position setup
9. Microscope Setup
10. Test tube minimum sample level teaching
11. Routine Operation

UriSed 3 PRO packaging content

1.
Main unit

2.
PC monitor

3.
PC unit
Driver discs
Windows installation disc
PC keyboard
PC mouse
Mouse pad



4.
Rotor
2-meter mains cable
USB cable
UTP cable
Leveler
12-millimeter graduated sample test tube
Cable ties
Water filter including a W2 tube
User Manual CD
Front cuvette rail
Back cuvette rail
Rotor fixing

9.
One hundred (100) 12-millimeter test tubes
One hundred (100) test tube stoppers
One hundred (100) rack adapter clips

5.
5-liter waste liquid container with the DIR and GRA tubes
5-liter distilled water container with the W3 tube
Liquid container stand

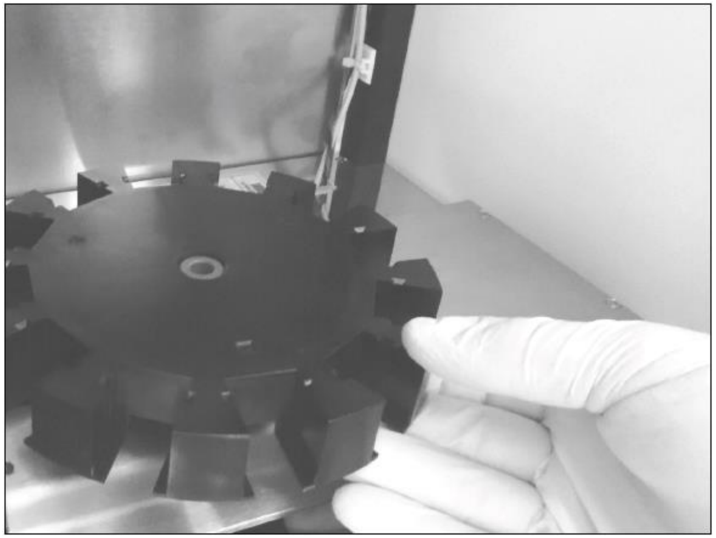
8.
Rack conveyor

6.
Ten (10) test tube racks with a capacity
of ten (10) test tubes each

7.
The Urinary Sediment by UriSed Technology' sediment atlas

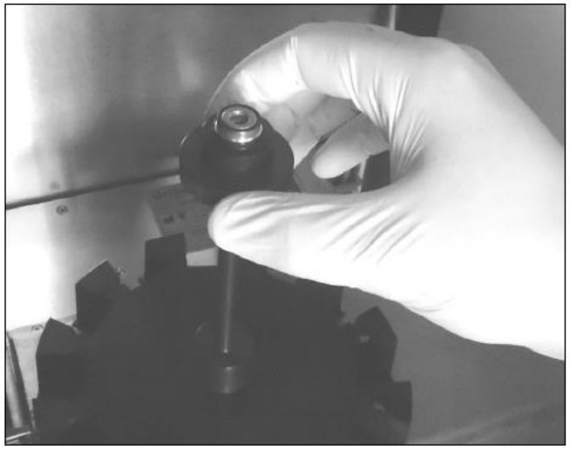
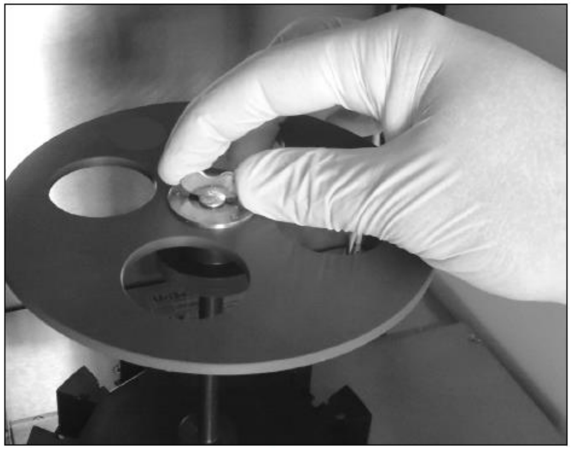
UriSed 3 PRO installation I.

1. Open the instrument doors and unscrew the retainer bolt that secures the robot during transportation.



2. Place the rotor on the pin of the axis holder.

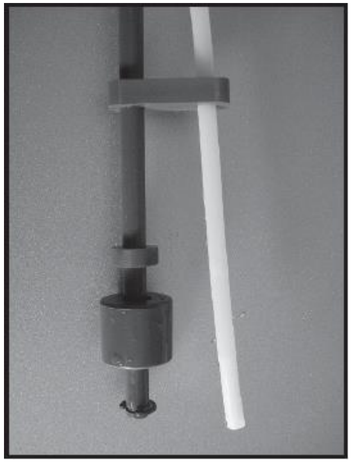
UriSed 3 PRO installation II.

3. Screw the rotor fixing tight into the pin to secure the rotor (see left).
4. Fit the feeder top disk on the rotor fixing and secure it with the mounted wing nut.


5. Place the front (see left) and back cuvette rail on.

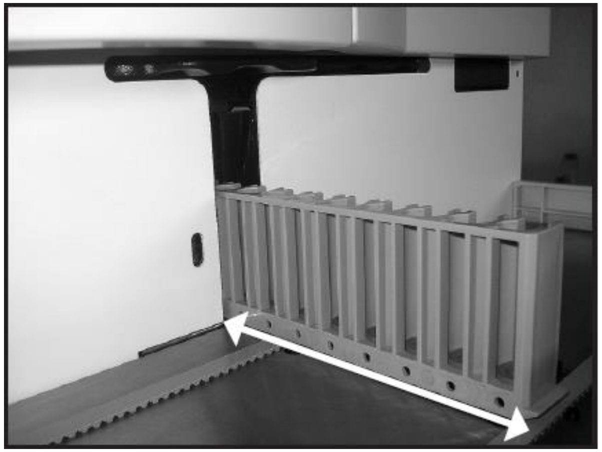
UriSed 3 PRO installation III.



Push the tube into the tank so that the 'Push until' label on the tube is exactly above the flexible tube guide on the cap of the tank.

Make sure that the end of the tube extends beyond the float switch rod.

Make sure that when the liquid tanks are in their final location, the gravity line tube is dropping all the way to the Waste tank. Avoid having the gravity tube double back upon itself.

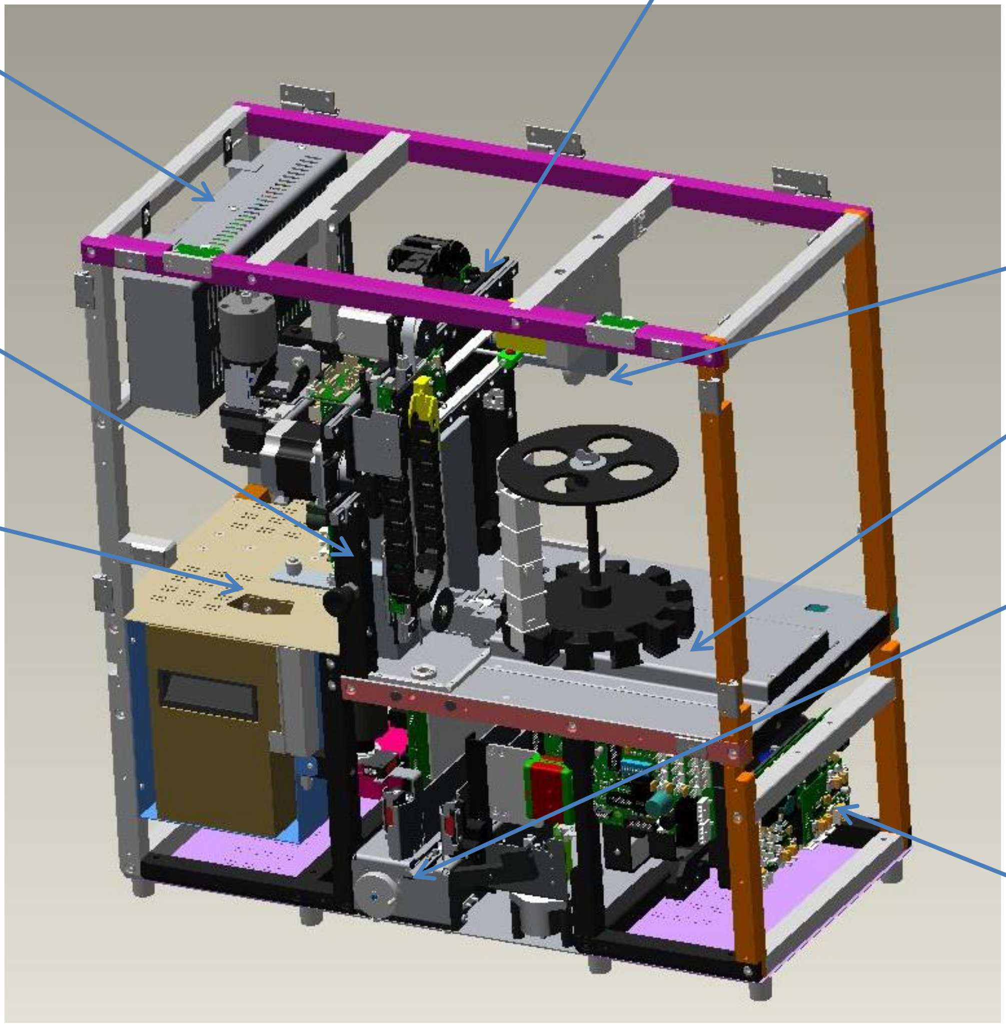


Place a test tube rack on the conveyor belts in the position where it is drawn into the device during normal operation.

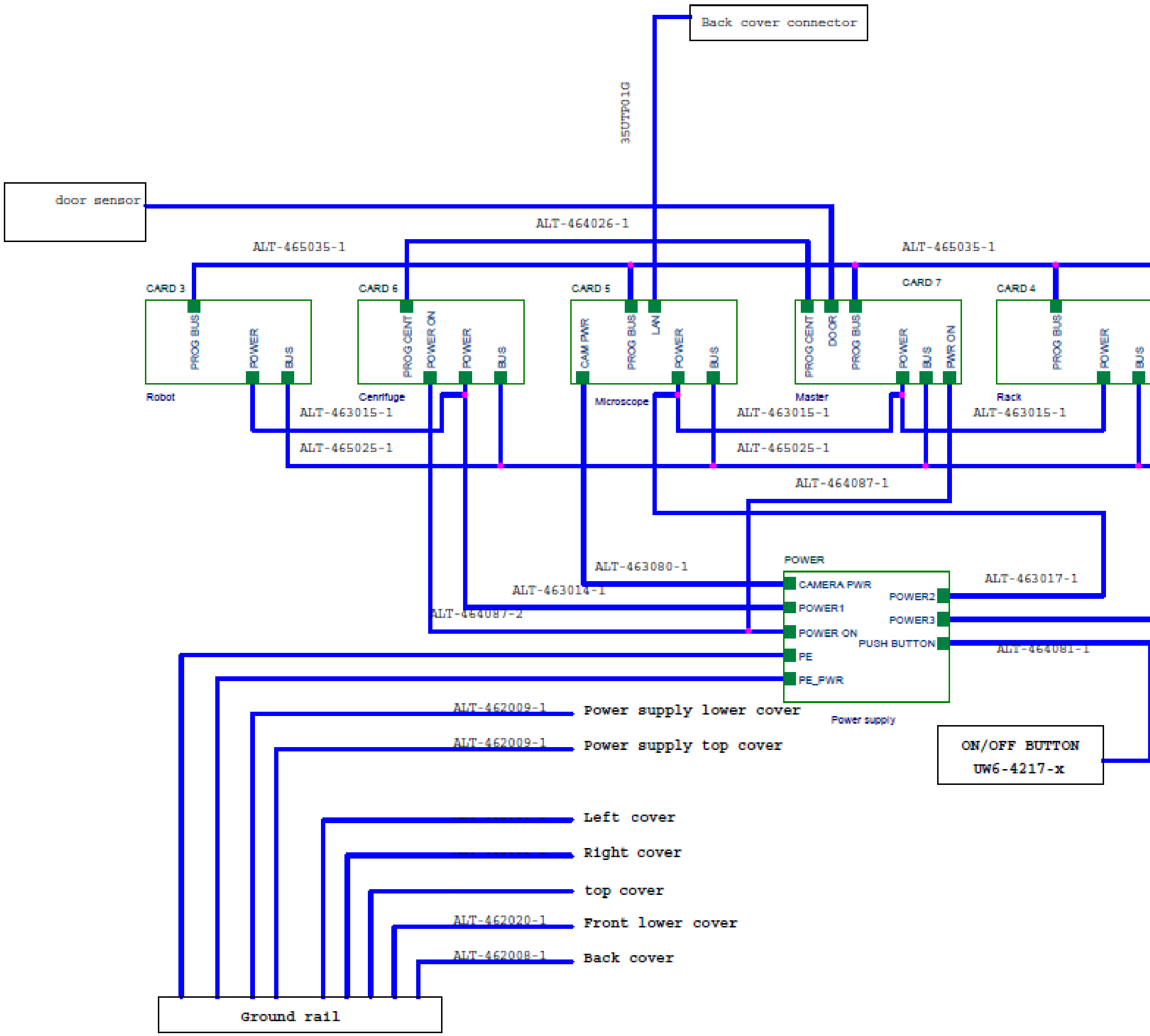
Make sure that the rack conveyor is flush with the front casing panel of the device.

The test tube rack should be able to be drawn into and pushed out of the device without any tilting or jerking.

Overview of the main modules



Block diagram



The power supply has two units:

- One generates 5V power for the non inductive sensors and PCBs
- Another generates 24V power for the inductive sensors and the motors
- The power supply also generates 12V power for the camera.

PCB and communication concept

PCBs with micro-step motion control

• One type for most of the motor controller boards including the feeder, washer, robot, rack and microscope boards – only the centrifuge and the master PCB are different

• Only one „empty” card is provided by 77e which has to be upgraded and the used controller type has to be defined when installing it into an instrument

Bus cable system

• One bus (RS485) cable for the normal communication and one extra bus cable (RS232) for the card programming

Advantages:
Super-fast upgrading of the cards.

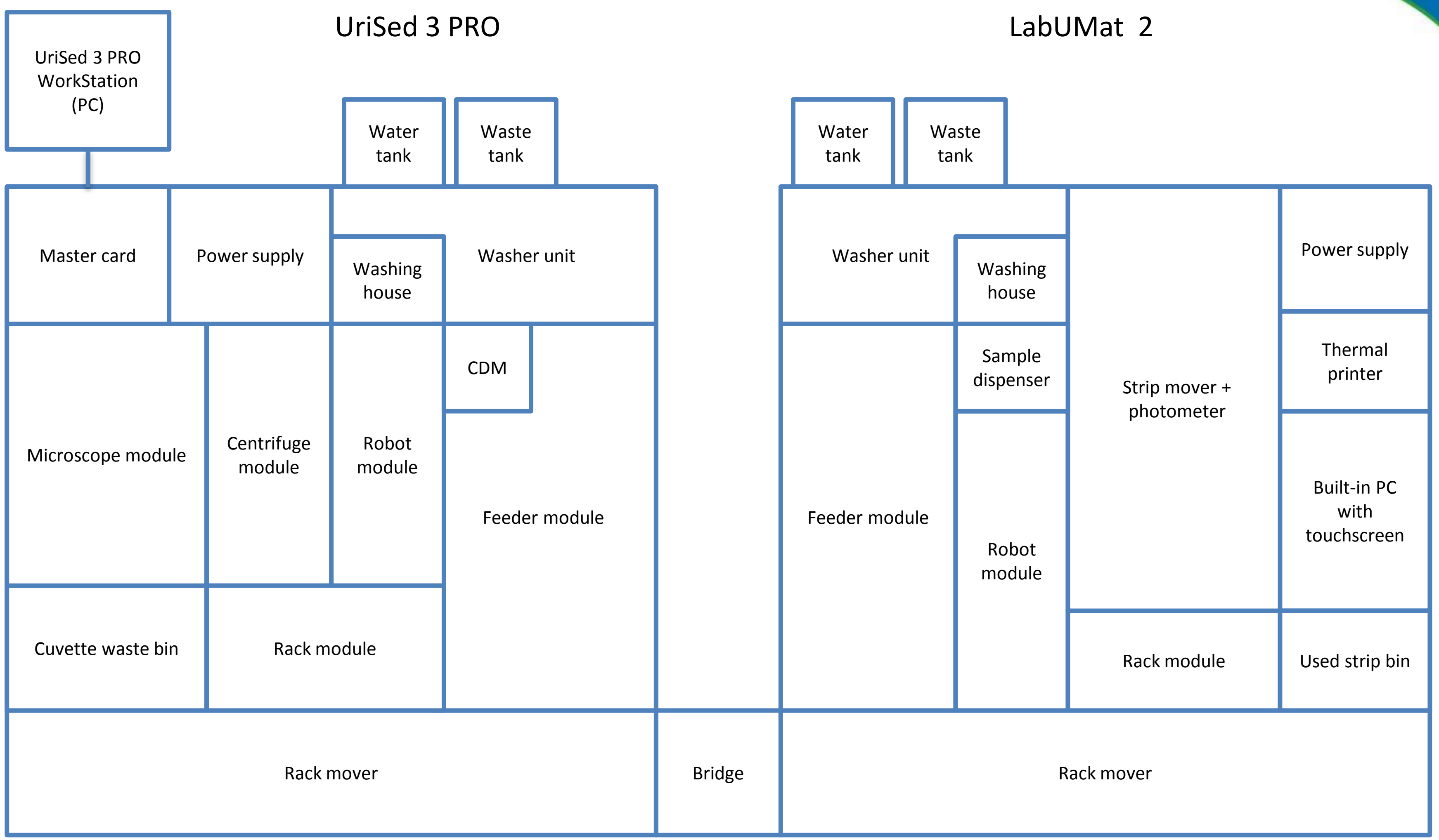
Interlock switch

• Two-way switch – one for the user and two for the service mode ☺

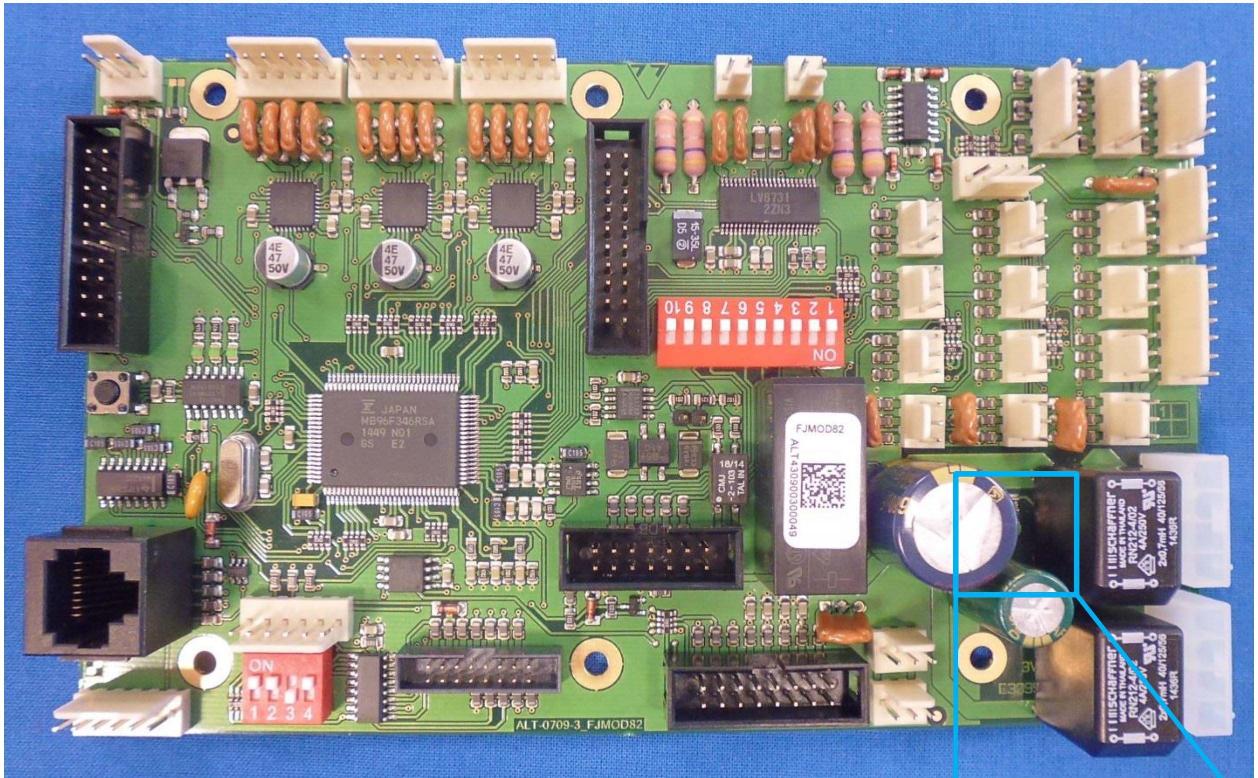
• The switch turns on and off the 24V but does not switch the 5V

Reason:
The door sensor has to be switched by hardware since they are hardware interlocks.

Modular structure of the connected systems



Motor controller board



There are 7 Boards in the System:

1. Feeder
2. Washer
3. Robot
4. Rack
5. Microscope
6. Centrifuge
7. Master

When ordering a PCB:

-PCB 1,2,3,4,5 are identical

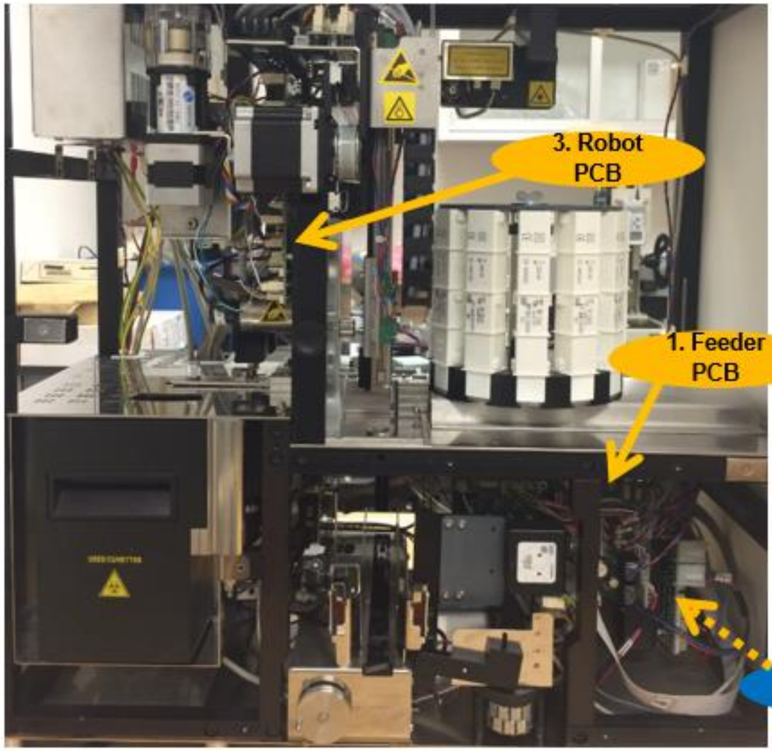
-PCB 6 and 7 are different

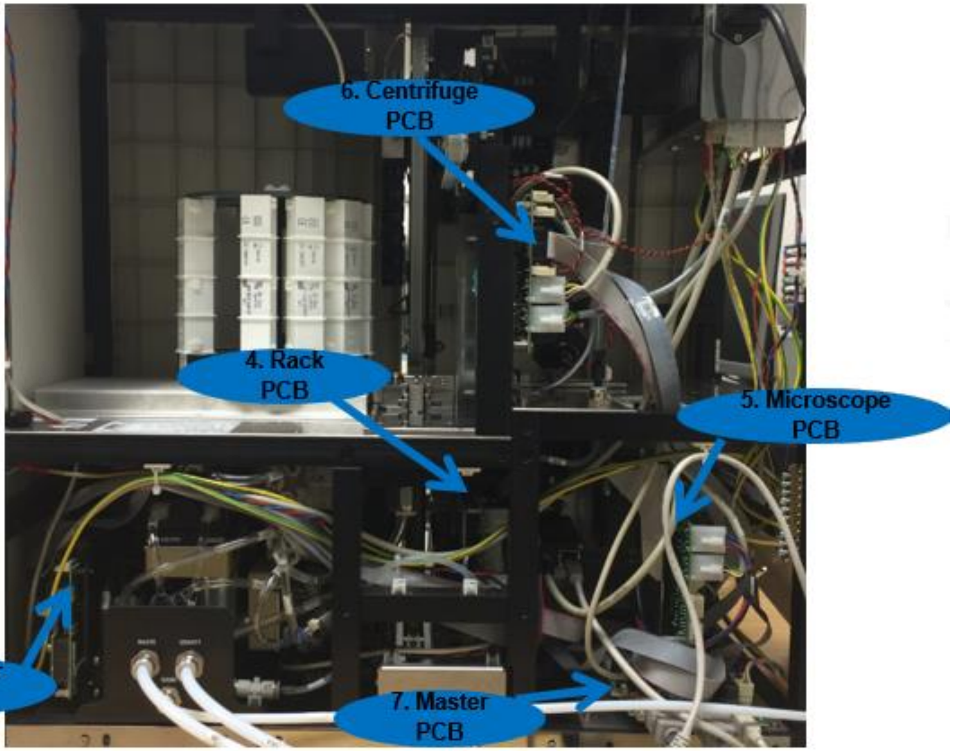
Power indicator LEDs:

PCB locations in the system

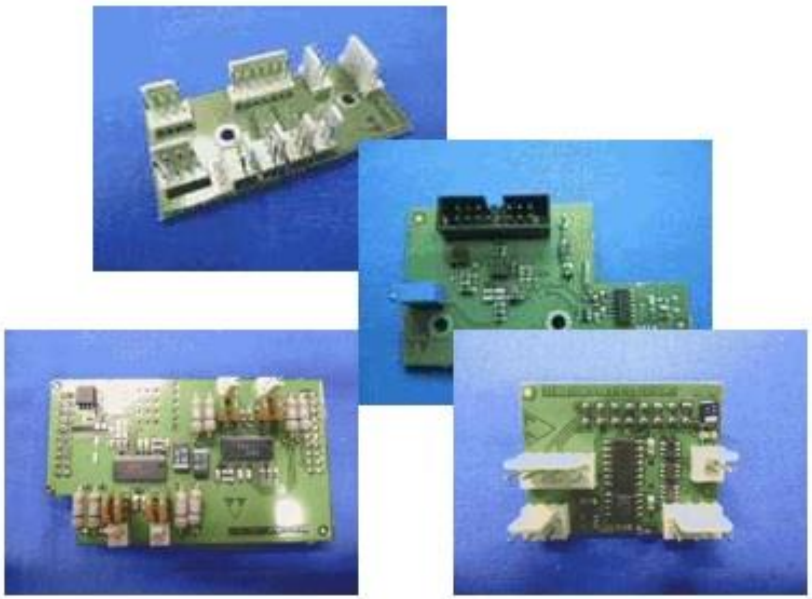
Front view of the system

Back view of the system





Other boards



- CDM interface Board (Feeder Module)

- Vertical Motor Interface Board (Robot Module)
- Oscillator PCB (Robot Module)

- Rack aligner Interface Board (Rack Module)
- Level Detector Interface Board (Rack Module)

- Washer extension board
- LED Driver Board

Motor controller board replacement

1. Configure the DIP switch and the jumper according to the jumper table
2. Replace the PCB
3. Start the service software and upgrade the card’s software
4. Configure the card on the „Card configuration” page
    1. Set the card type
    2. Set the card version
    3. Then select „start configuration”
5. Set the instrument serial number
6. Use position setup if neccessary (pipetting module, feeder module)
7. Start User software

Additional information:

Each module has a white label on it with a series of 14 characters

-The first 9 characters are for the Module **part number**

-The last five character are for the module **serial number**

Circuit board DIP switch settings

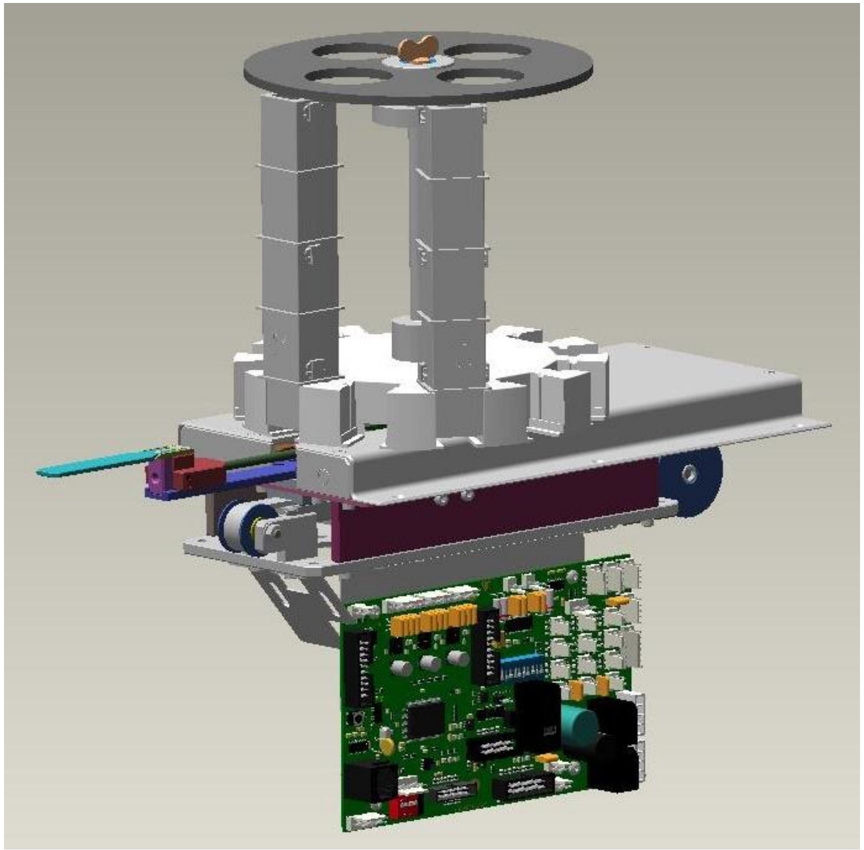
Module	Address	DIP switch	DIP „ON"	JUMPER PCB: FJMOD82	DIP „ON"
Cuvette feeder	1		2,3,4		-
Washer	2		1,3,4		-
Robot	3		3,4		10
Rack	4		1,2,4		10
Microscope	5		2,4		2,4,5,7
Centrifuge	6		1,4	n/a	n/a
Master	n/a	n/a	n/a	n/a	n/a

1. Cuvette feeder module

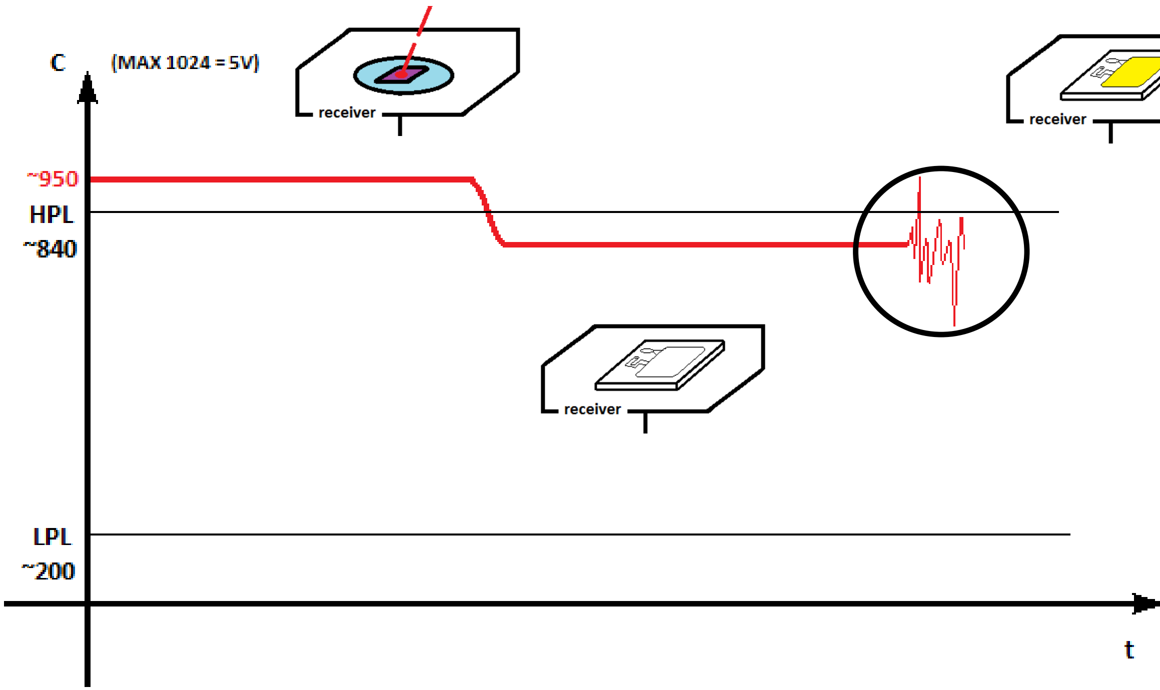
Main components
• pushing arm
• cuvette holder turning motor
• CDM (Cuvette Detection Module)

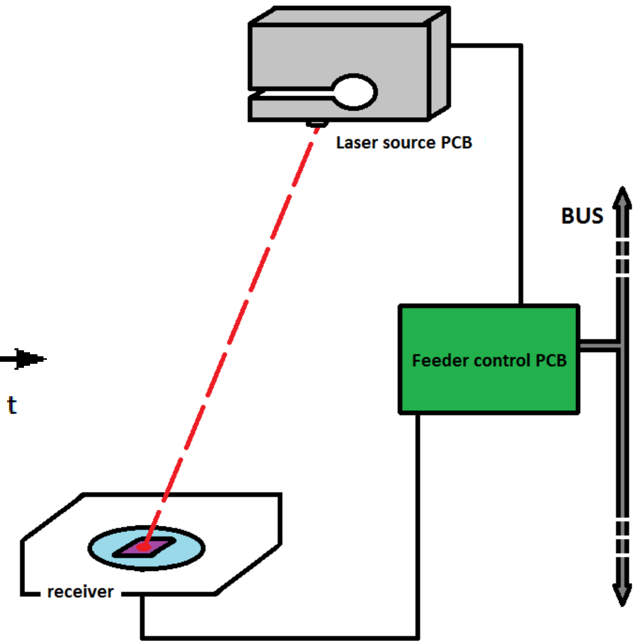
Functions
• holds the cuvette cartridges
• forwards the cuvettes one by one into specified positions with the help of the pushing arm
• from the cartridge to the microscope arm through pipetting and centrifuge positions

CDM components
• Laser Diode
• Phototransistor



1. CDM



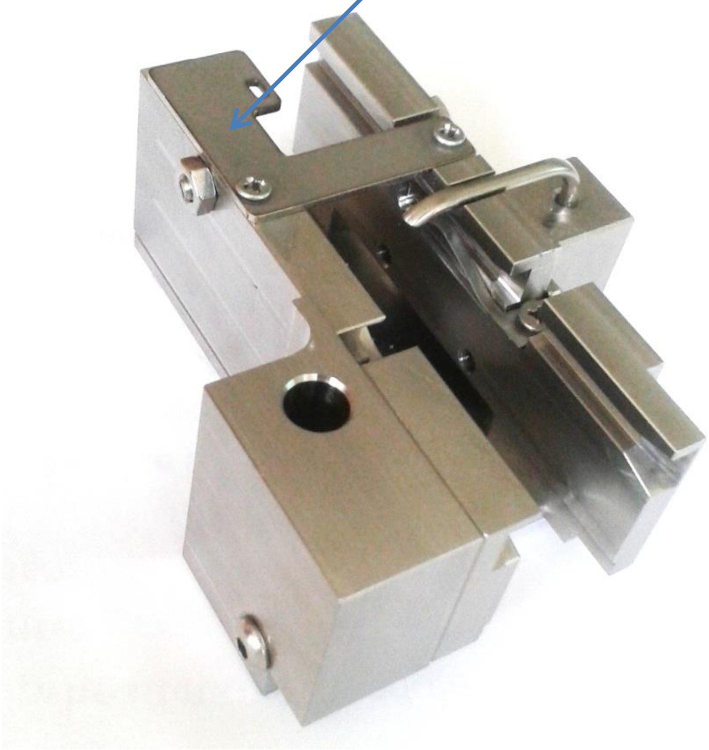


The laser beam is aimed at a specific point, passes through the cuvette chamber, reaches the photostransistor, a value is generated on the Sensors tab.
The Cuvette Detection Module senses if the cuvette was loaded and if the cuvette was properly filled up with urine.
The value is the highest if there is no cuvette present, and decreases as the cuvette gets filled up with fluid.

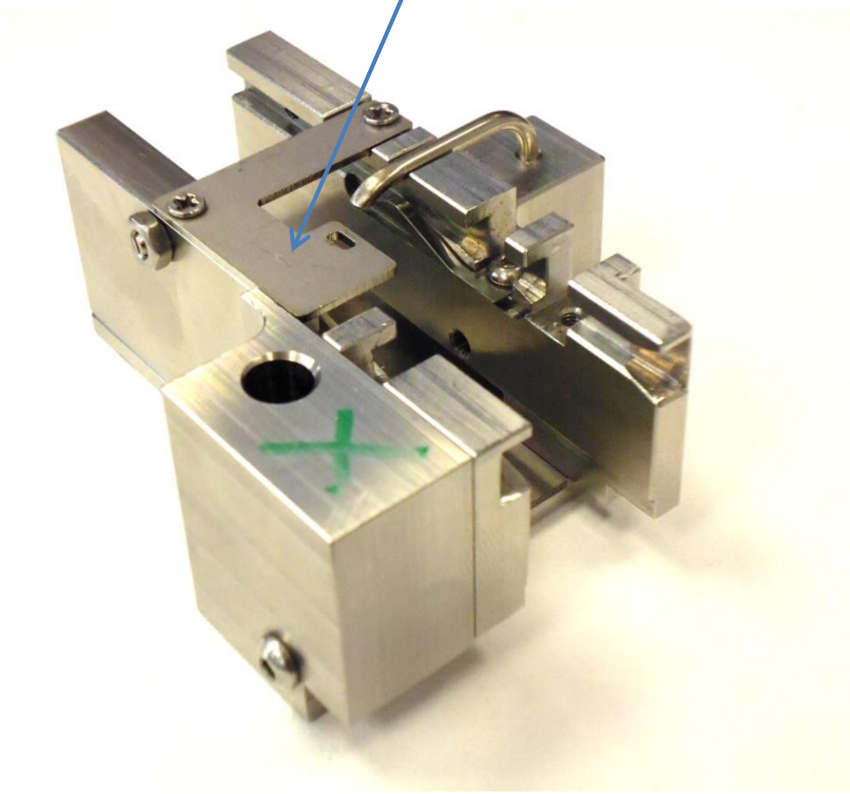
1. Feeder module

Front cuvette rail - CDM laser adjuster plate positions:

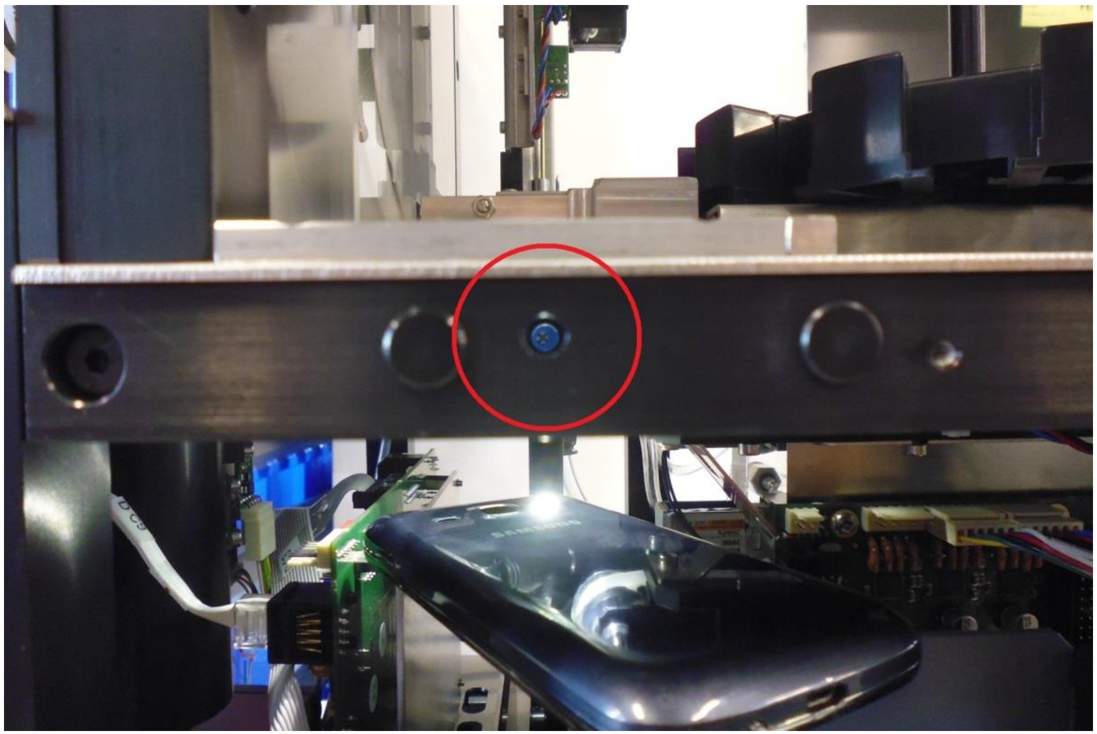
Normal position



Adjusting position



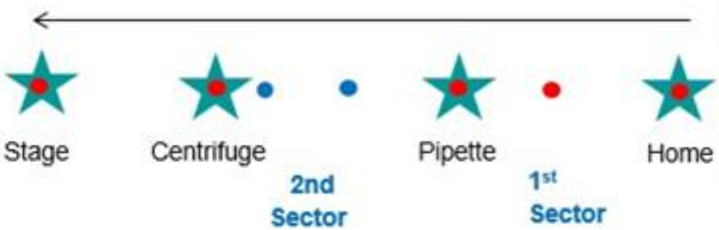
1. Adjusting the C(uvette) value

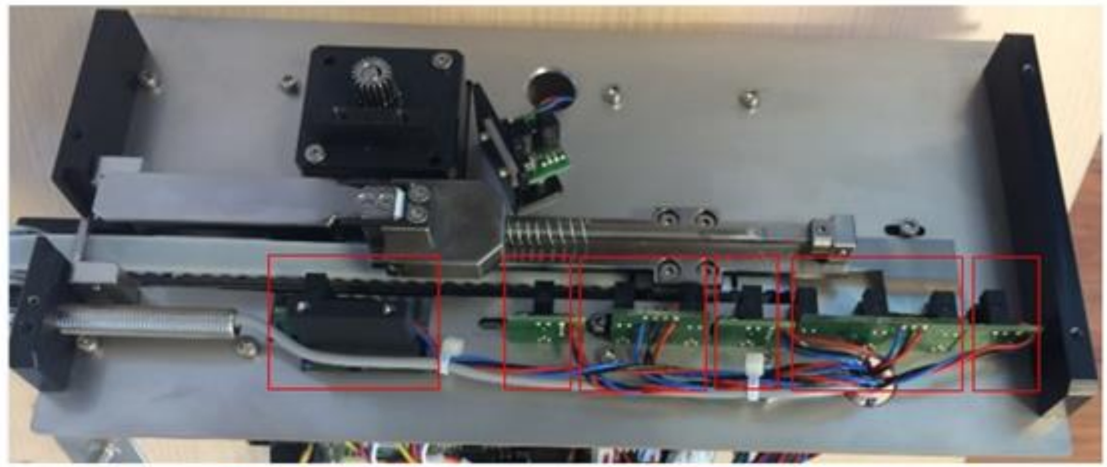


- Remove the front chassis panel
- Take the cuvette pushing arm to pipetting position to turn the laser on
- Navigate to the sensor test tab and check the C(uvette) value
- The C(uvette) value has to be 950 +- 10 without cuvette in the Front Guide.
- Turn the potcentimeter (in the hole circled red) until the C(uvette) value is 950 +-10 if it’s needed (Use a long Phillips screwdriver)

1. Cuvette Feeder Sensors

Pusher Arm Sensors
6 positions

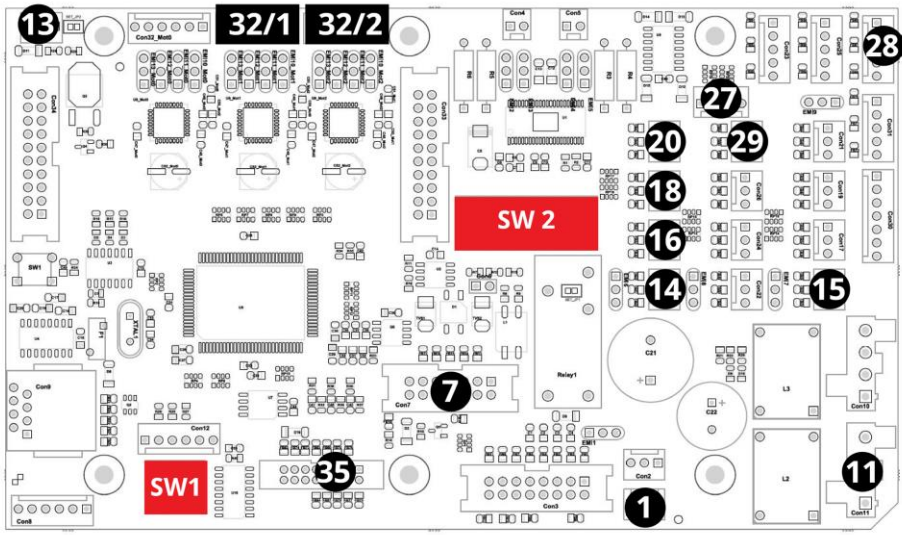




Stage	Centrifuge	2nd Sector	Pipette	1st Sector	Home	Sensor
20	18	27	16	15	14	Conn on PCB 1
OPTO ASSY L350	OPTO ASSY L350	OPTO DUAL ASSY L200	OPTO ASSY L200	OPTO TRIPLE ASSY L200	OPTO ASSY L200	Sensor Type

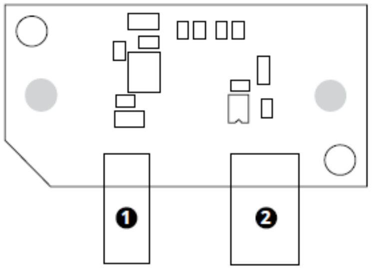
Active after initialization

1. Cuvette feeder module



PCB layout

1 Cuvette presence and fillup sensor connector
7 Bus cable connector
11 Power cable connector coming from PCB 2 (Washer module)
13 CDM interface board connector
14 Pushing arm home position sensor and connector
15 1st sector sector sensor and connector
16 Pushing arm pipette position sensor and connector
18 Pushing arm centrifuge position sensor and connector
20 Pushing arm microscope position sensor and connector
27 2nd sector sensor and connector
28 Front cuvette rail presence sensor and connector
29 Rotor sensor and connector
32/1 Pushing arm motor connector
32/2 Rotor motor connector
35 Programming bus cable connector
SW1 SW1 - PCB address DIP switch (only slider 1 is in the down position)
SW2 SW2 - configuration DIP switch

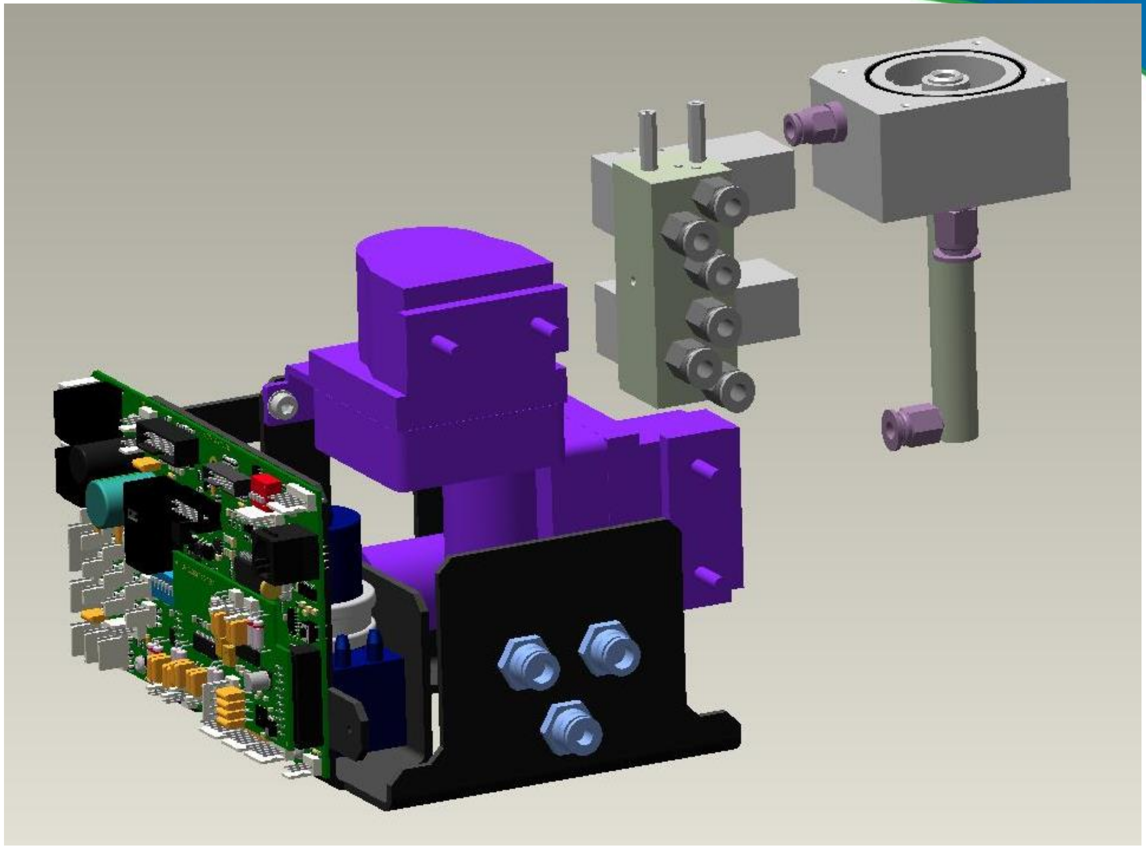


1 Laser diode connector
2 Circuit board number 1 board-to-board connector

2. Washer module

Main components

- Inner washer peristaltic pump
- Outer washer peristaltic pump
- Membrane suction pump
- Aspiration pump
- Suction stub
- 2 electromagnetic valves



Functions

- Rinsing the pipette between sampletaking
- Regular disinfection of the system
- Draining excessive urine from the cuvette's surface
- Performing sample aspiration and pipetting

Inner Washing pump: *Water from washing tank into rinse well through Pipette*

Outer washing Pump: *Water from bottle into rinse well directly and stub*

Waste Pump: *Drains into the waste tank*

Aspiration Pump*: Mixes urine sample and Aspirates it into pipette*

Suction Stub: *Sucks excess urine from cuvette into Waste*

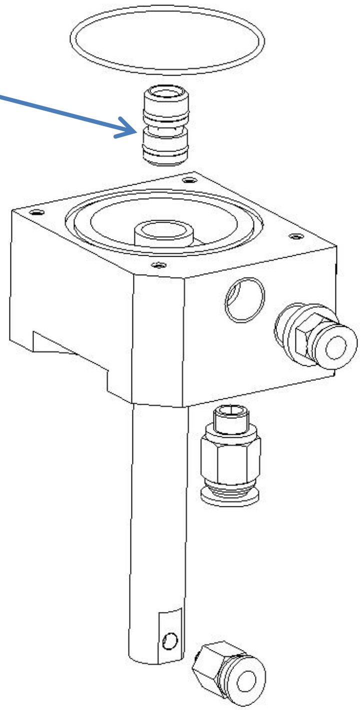
Electromagnetic Valves: *Direct flow of water and waste to and from stub and rinse well.*

2. Washer module

Washing house

• Washing of the probe through six small holes from the side

Advantages:
The probe washing station allows for an efficient cleaning of the sample probe.



Filters

Advantages:
Filtration to eliminate bubbles in the water line and to improve the lifetime of the suction pump.

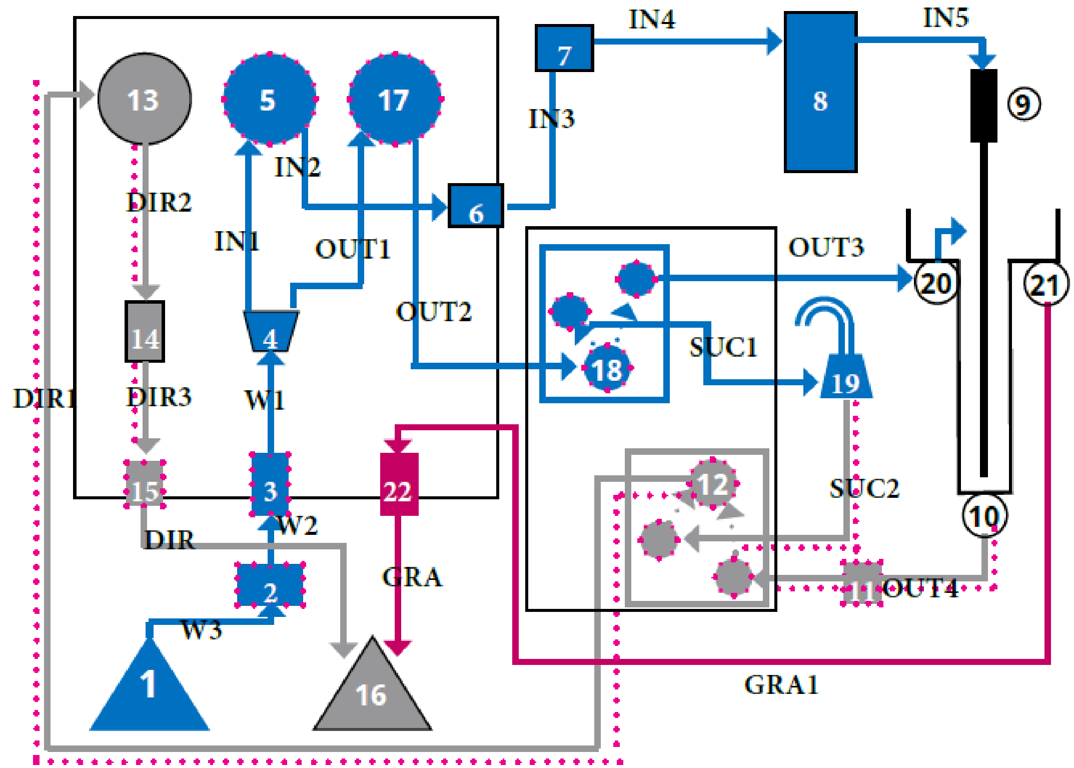
Interface PCB board

• „small" PCB to drive the membrane suction pump and the washer and waste valves.

2. Washer module tubing diagram

Main components

2. Water filter
5. Inner washer peristaltic pump
8. Aspiration pump with valve
9. Pipette
11. Water filter
12. Waste valve
13. Suction / membrane pump
17. Outer washer peristaltic pump
18. IF water valve
19. Suction stub



the IF water line

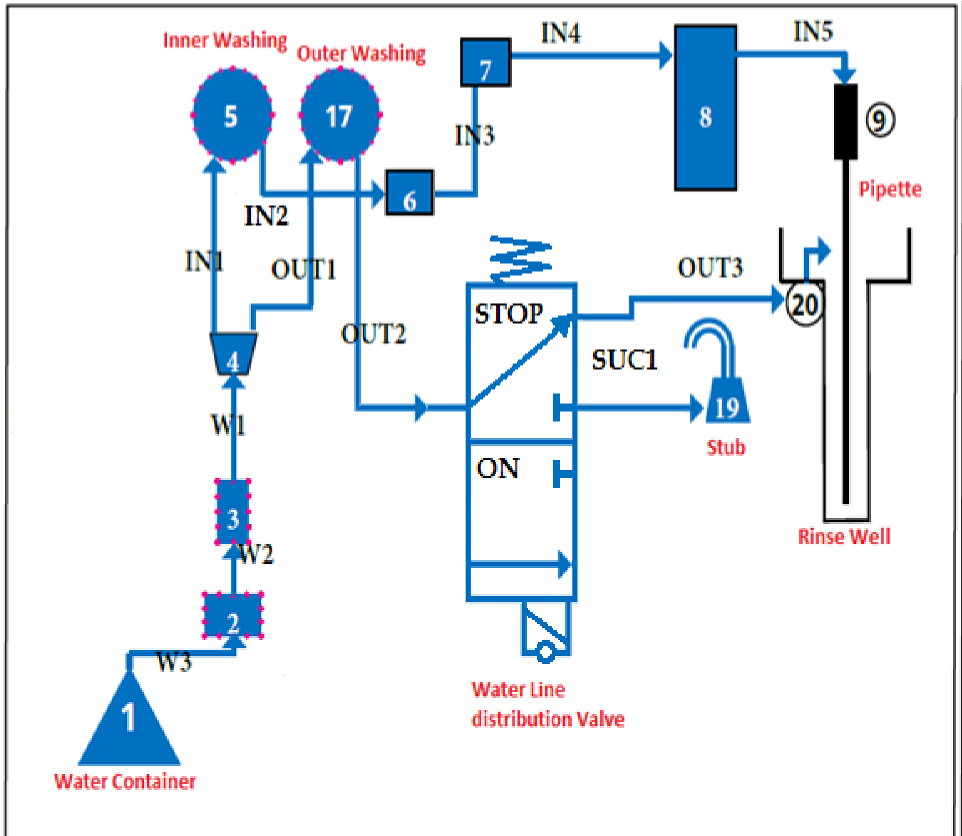
the gravity line

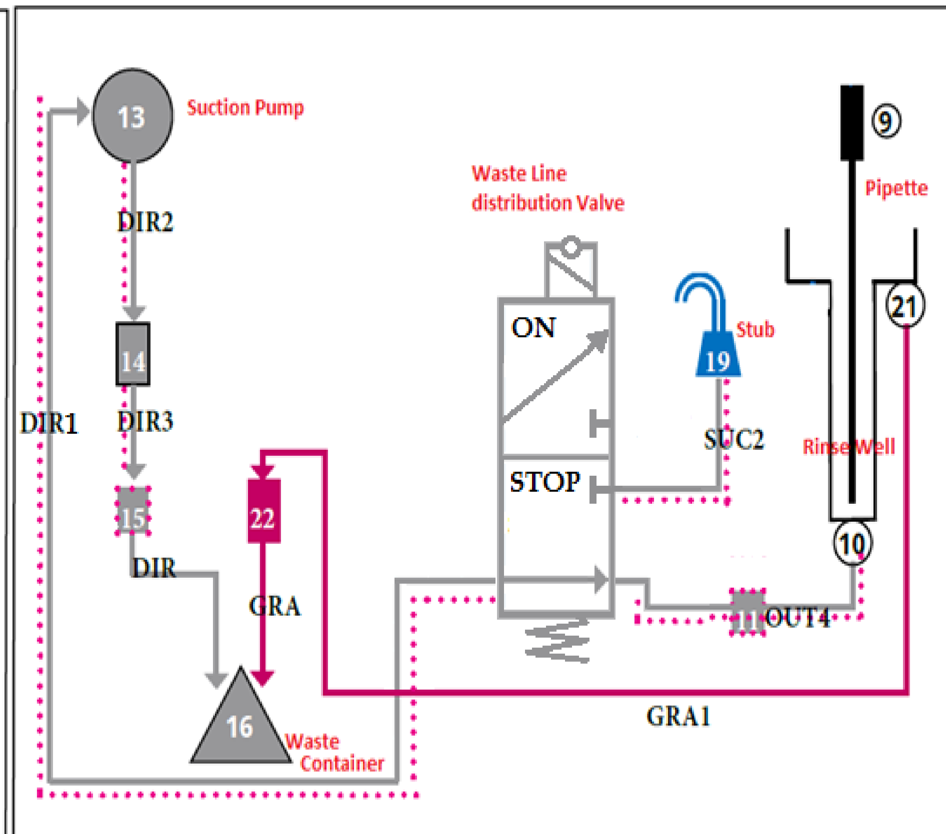
the waste line

Yearly maintenance up to 300 samples per day:

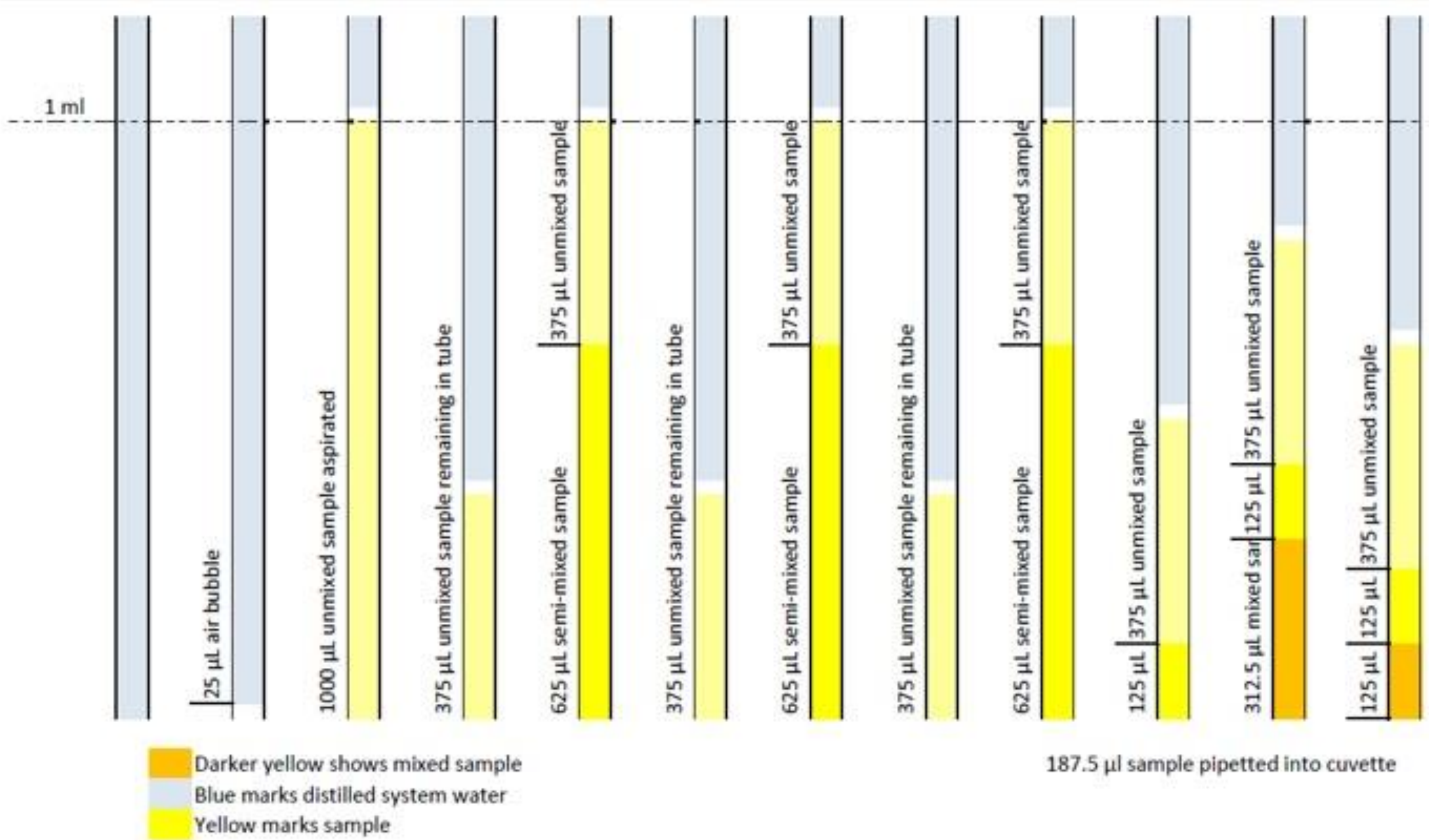
- *Waste line tubing (marked by gray)*
- Neoprene tubing of the peristaltic pumps

2. Washer module tubing diagram 2.

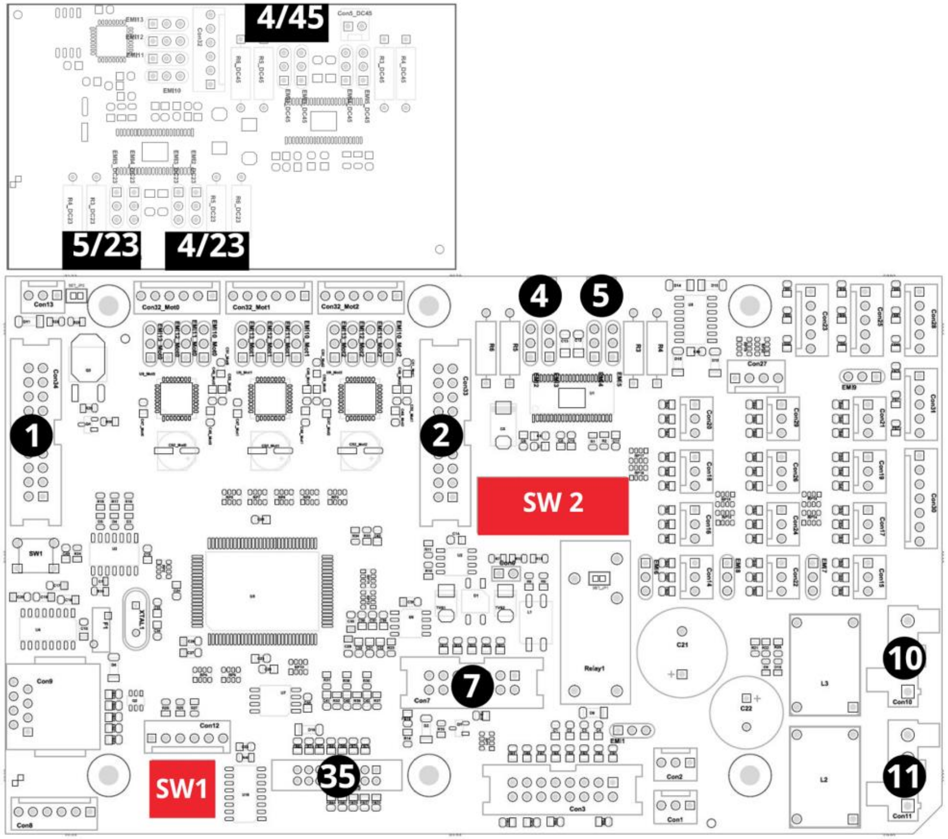




2. The mixing process



2. Washer module PCB



PCB layout

1 Extension board connector
2 Extension board connector
4 Inner washing (pipette rinsing) pump motor connector
5 Outer washing (washing house fillup) pump motor connector
7 Bus cable connector
10 Power connector to Feeder PCB
11 Power connector
35 Programming bus cable connector
SW1 SW1 - PCB address DIP switch (only slider 2 is in the down position)
SW2 SW2 - Configuration DIP switch (with all sliders off)
4/45 Waste pump motor connector [on the extention card]
4/23 Water line distribution (outer washer) valve connector [on the extention card]
5/23 Waste line distribution (suction) valve connector [on the extention card]

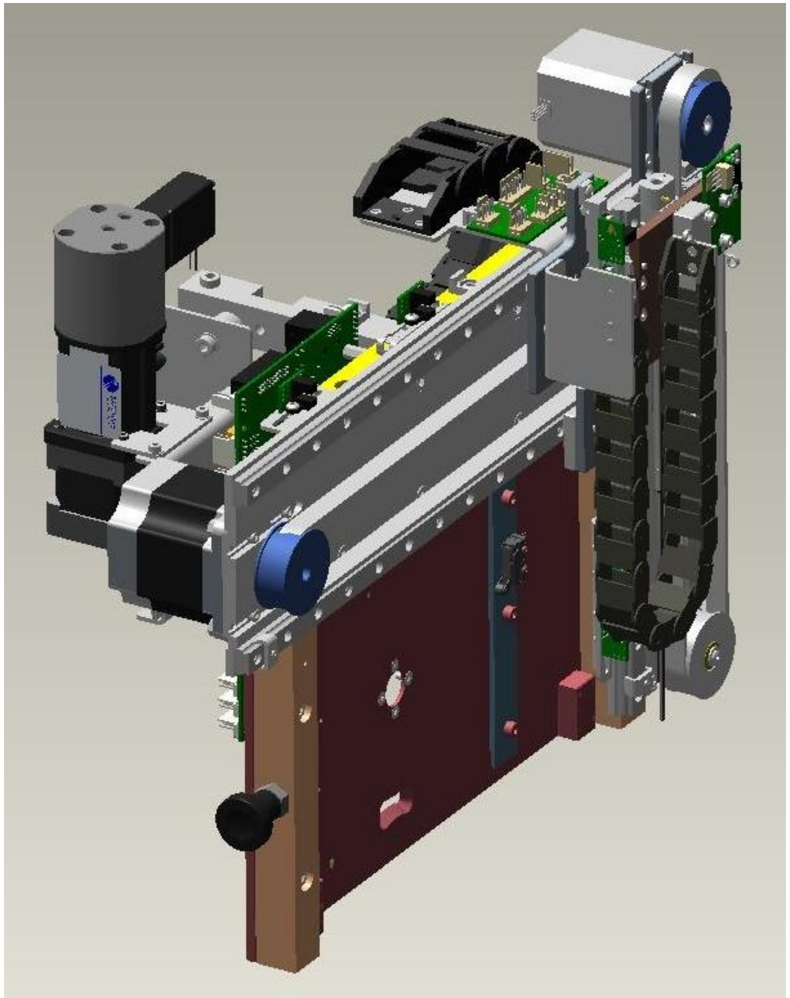
3. Robot module

Main components
- Y (up-down) motor
- X (back-front) motor
- Pipette
- Aspiration pump

Functions
- Sample aspiration and mixing(needle)
- Sample level detection (serves as an antenna)

Vertical pipette positions
- Up
- Pipetting
- Washer
- Tube

Horizontal pipette positions
- Pipetting/cuvette fillup
- Tube/sample aspiration
- Washer (home)

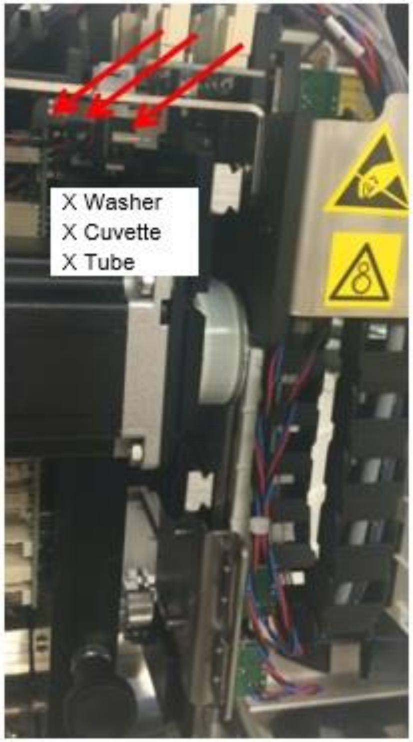


3. Robot module sensors

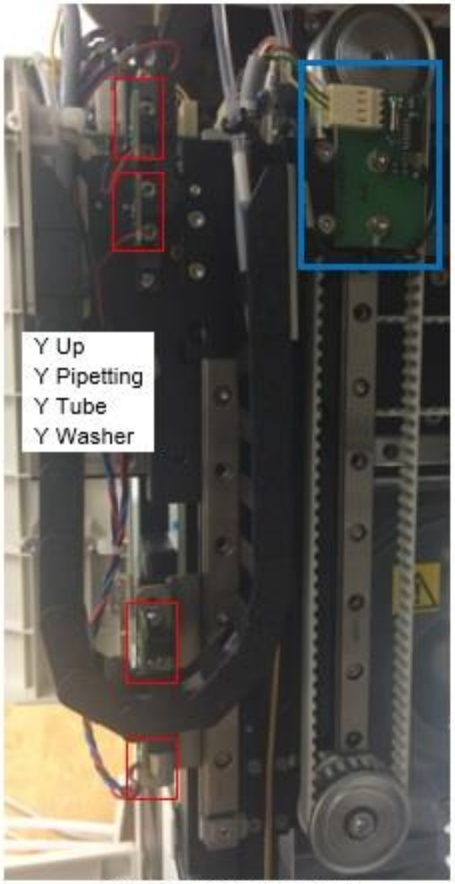
Horizontal Rinse well position sensor
-> Robot X washer
Horizontal cuvette fillup position sensor
-> Robot X Cuvette
Horizontal test tube position sensor
-> Robot X Tube

Pipette up position sensor
-> Robot Y Up
Vertical cuvette fillup position sensor
-> Robot Y Pipette
Lower end position sensor:
-> Robot Y Tube
Vertical rinse well position sensor
-> Robot Y washer

Liquid level sensor
(not a physical sensor, PCB can be replaced)
-> Oscillator PCB plugged into PCB 3
-> Sensor PCB plugged into PCB 4

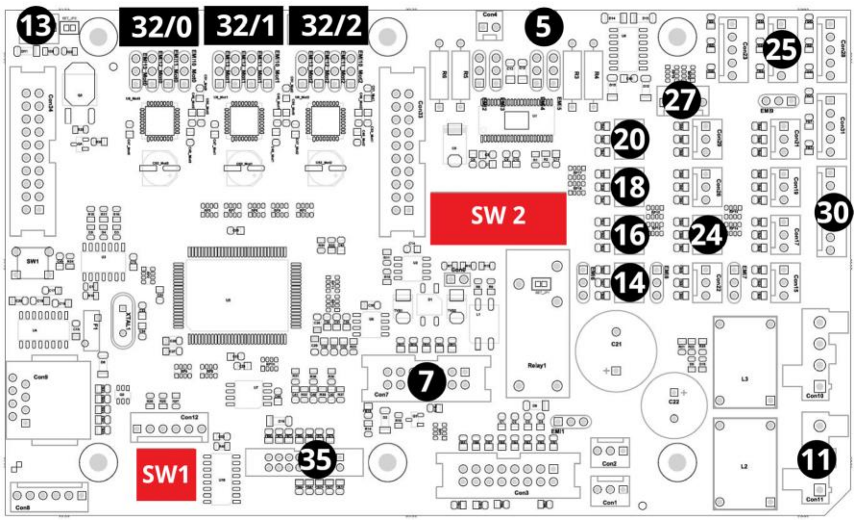


Side view from the left



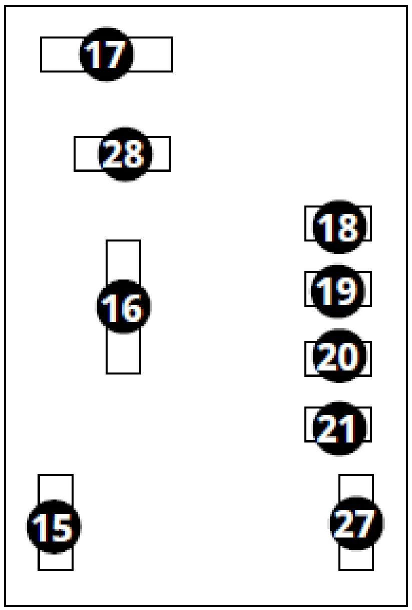
Side view from the right

3. Robot module PCB



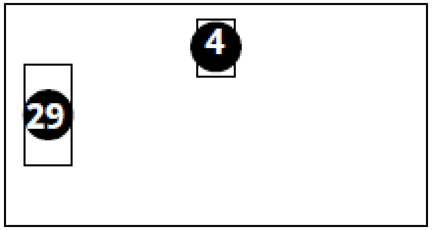
PCB layout
5 Aspiration valve connector
7 Bus cable connector
11 Power connector (to PCB 6)
13 Aspiration pump LED power (red/black)
14 Robot X washer position sensor connector
16 Robot X pipette position sensor connector
18 Robot X tube position sensor connector
20 Oscillator PCB board connector
25 Oscillator PCB 24-volt connector
24 NC
27 Aspirating pump home position sensor connector (in case of 2.5 ml pump)
30 Robot Y (Vertical motor) interface board sensor connector
32/0 Robot Y (Vertical motor) interface board
32/1 Robot X (Horizontal motor) connector
32/2 Aspirating pump motor connector
35 Programming bus cable connector
SW1 PCB address DIP switch (sliders 1 and 2 are in the down position)
SW2 Configuration DIP switches (with only slider 10 on)

3. Robot module PCB 2



15 Oscillator power connector
16 PCB 3 sensor connector
17 Robot vertical motor connector
18 Robot Y up position sensor connector
19 Robot Y pipette position sensor connector

20 Robot Y wash sensor connector
21 Robot Y tube position sensor connector
27 Oscillator PCB board connector
28 PCB 3 motor connector



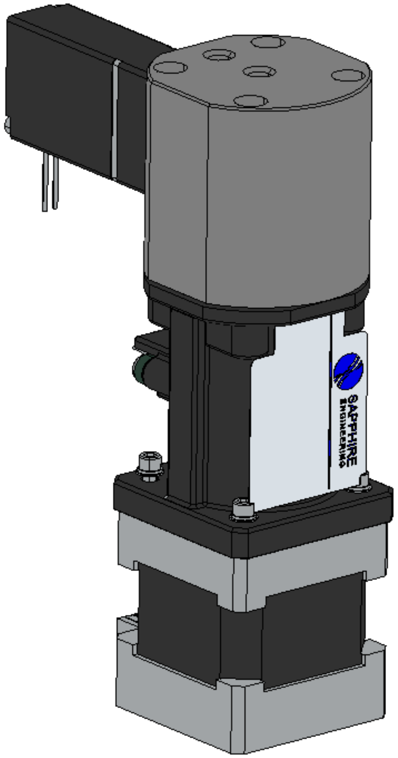
29 Vertical motor interface board connector
4 Pipette signal

3. Aspiration pump

Sample pump unit

• Micro-step controlled aspiration pump unit with valve with precisely controlled motions

Advantages:
Silent, does not require yearly maintenance and lasts for the lifetime of the instrument.

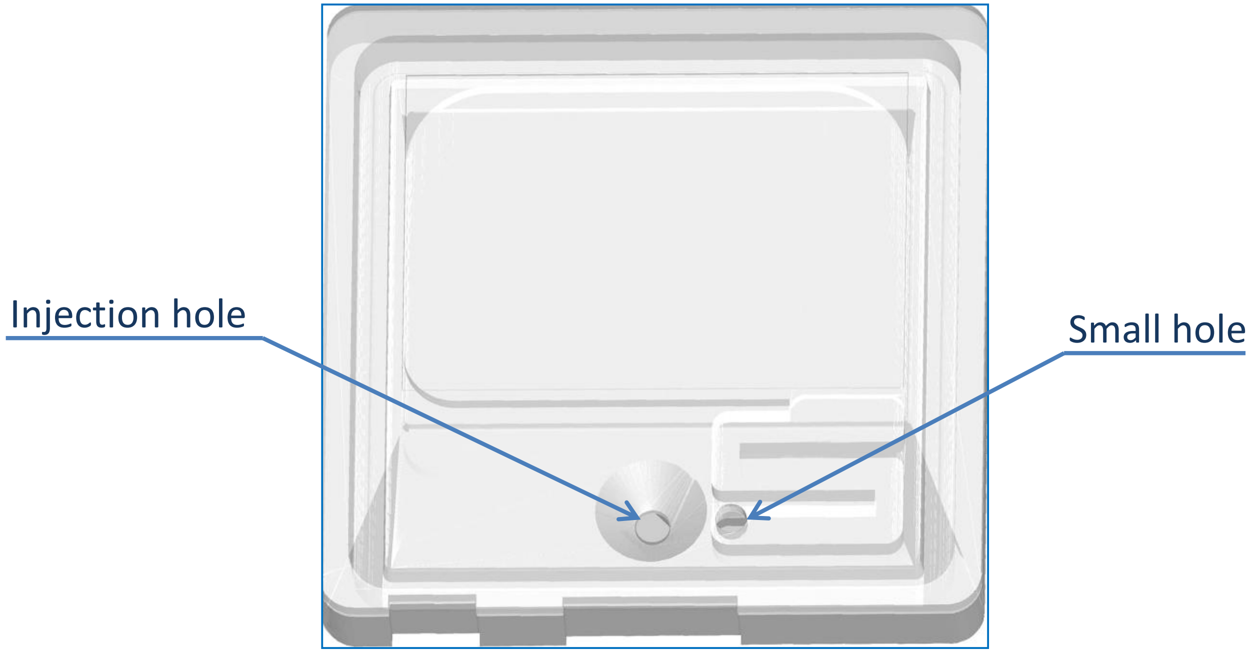


Fill-up sequence

Cuvette fill-up sequence

- The pipette goes into the injection hole and fills up the cuvette with ~187,5 µl of urine
- The pipette comes out from the injection hole and sucks the overflowed urine above it
- The pipette moves to the small hole at the meander and sucks the overflowed urine

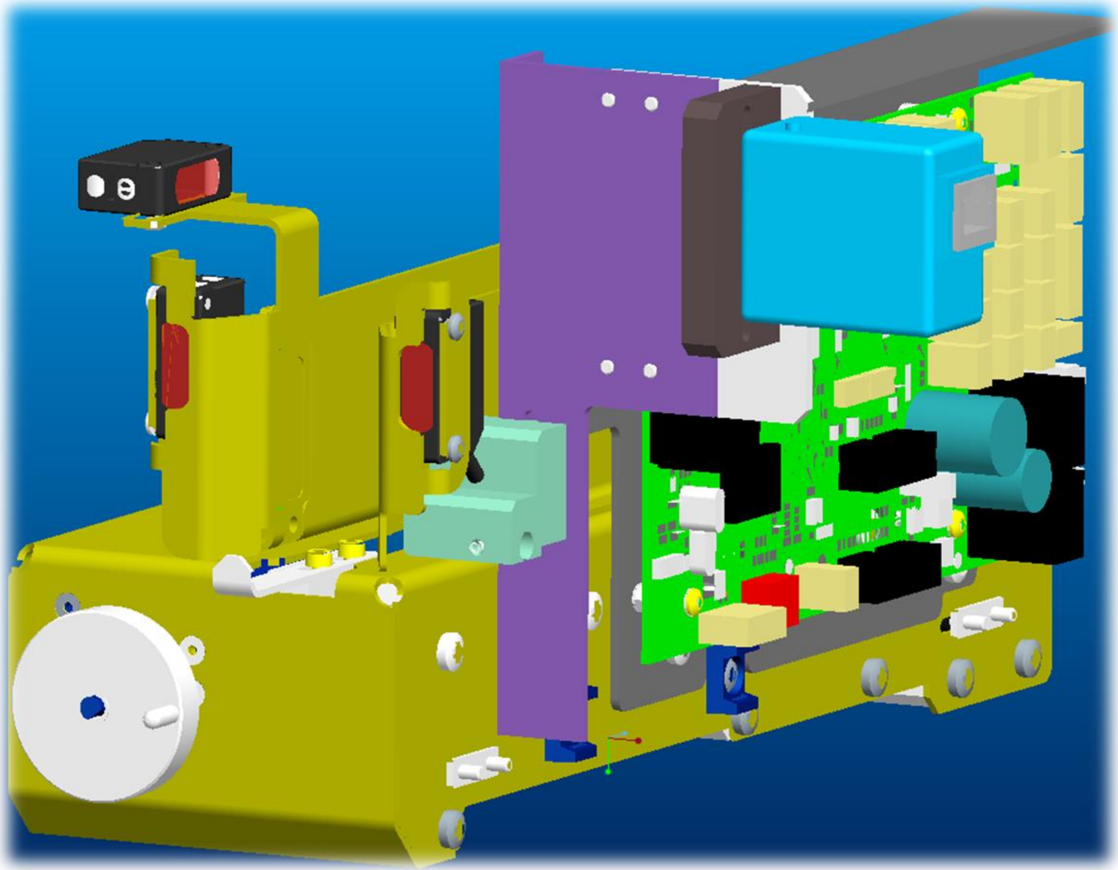
Advantages:
With the suction step – over the small hole – it is ensured that no sample will be present on the surface of the cuvette during picture taking.



4. Rack module

Main components

- Rack aligner module
- Conveyor motor (DC motor)
- Rack mover motor
- Barcode reader
- Tube sensor
- Rack arrive and full sensors
- Sample level detector.



Function

- Rack traonsportation
- Rack positioning

Module description:
The Rack module is responsible for the rack transportation and positioning.

4. Rack module sensors

SENSORS

Rack Delivery Sensor:
- Photoelectric sensor
- Rack is present to move to passage

Conveyor Max capacity sensor:
- Photoelectric sensor

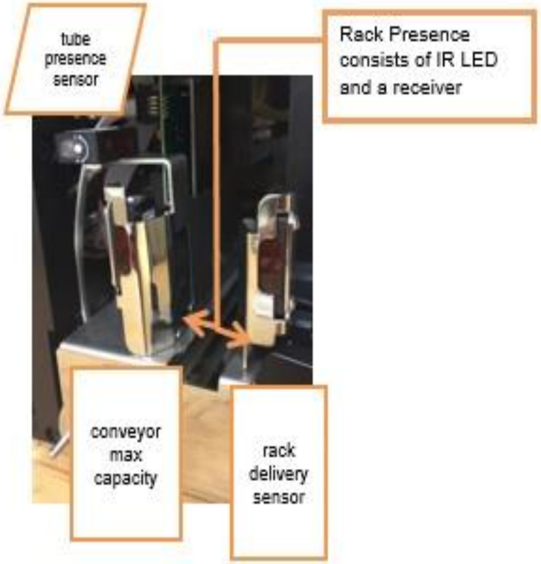
Tube presence sensor:
- Photoelectric sensor
- Tube present in the hole under test tube aperture

Rack Position sensor (no physical sensor, receiver on Level detector board and/or IR LED may be replaced)
- Passive IR LED proximity sensor
- Nothing is blocking the sensor beam (either passage is clear or there is no tube on rack below the tube aperture. Check **Rack Hole** value

Rack aligner sensor x2 (part of rack aligner board):
- Transmissive Photo interrupter
- Open – Close

Rack puller sensors x2:
- Transmissive Photo interrupter
- Rack in – rack out

Liquid level sensor:
- Receiver of oscillation signal by pipette, check **Level** value



RACK ALIGNER INTERFACE BOARD WITH THE 2 RACK ALIGNER SENSORS ATTACHED

Rack module hardware improvements

Barcode reader
(RS232)

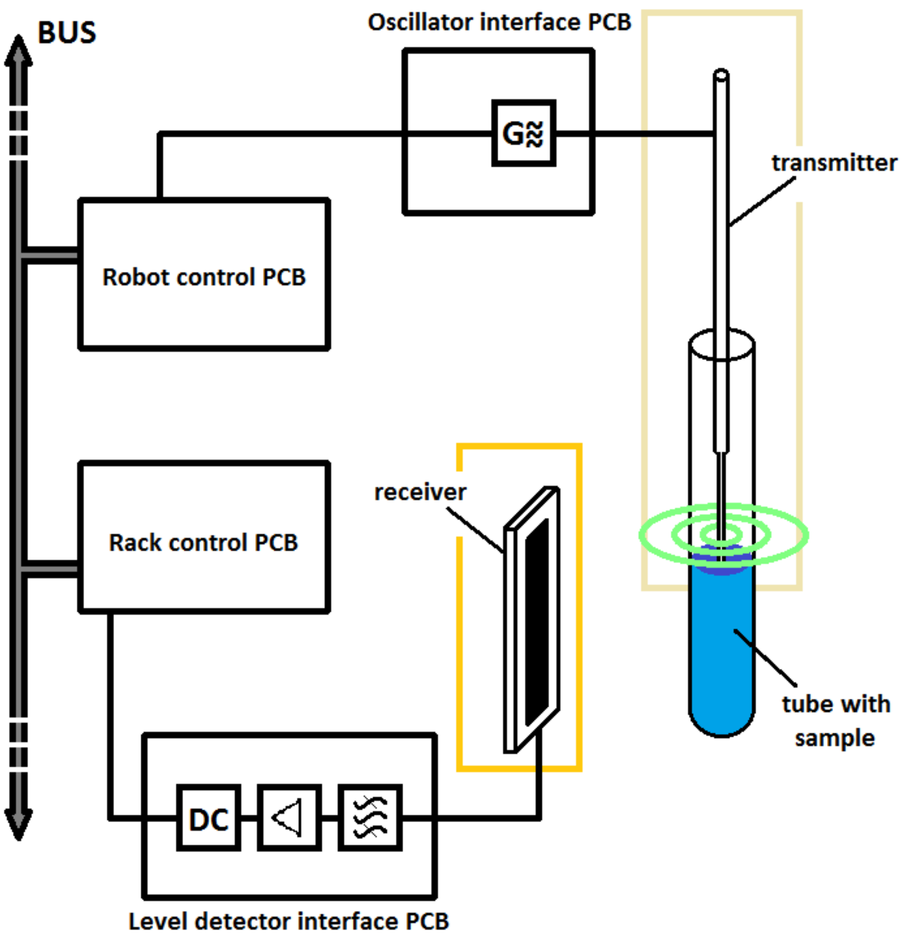
•JADAK FM-5 type

The reader is capable to „focus" only a specific area (window) by software -> no need of the diaphragm.

This barcode reader supports the following barcodes:

Linear:	UPC/EAN/JAN, GS1 DataBar, Code 39, Code 128, Code 32, Code 93, Codabar/NW7, Interleaved 2 of 5, Code 2 of 5, Matrix 2 of 5, MSI, Telepen, Trioptic, China Post; 2D Stacked: PDF417, MicroPDF417, GS1 Composite
2D Matrix:	Aztec Code, Data Matrix, QR Code, Micro QR Code, MaxiCode, Han Xin Code
Postal:	Intelligent Mail Barcode, Postal-4i, Australian Post, British Post, Canadian Post, Japanese Post, Netherlands (KIX) Post, Postnet, Planet Code;
OCR:	OCR-A, OCR-B, E13B (MICR) (OCR Optional)

Level detection



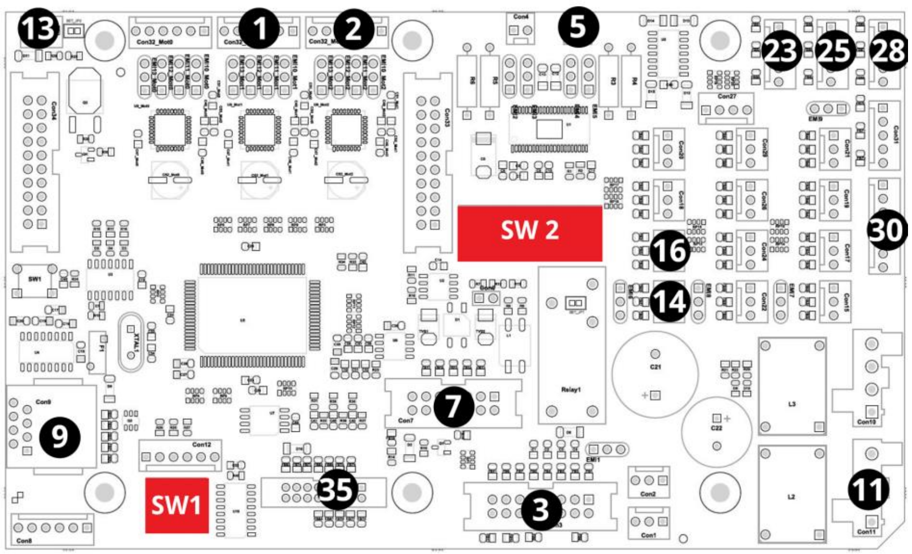
- Based on an oscillator circuit; a 32 kHz signal is transmitted and the signal change will be detected
- The probe is the transmitter of this 32 kHz signal
- A silver metal plate on the side of the rack module is the receiver
- A change in the signal is detected once the probe encounters the sample surface
- The position of the probe at the sample surface is now determined and the mixing steps are defined accordingly

- Sample volume: min. 2.0 ml
- Detected values:

  Probe is in sample ~630

  Probe is out of sample ~480

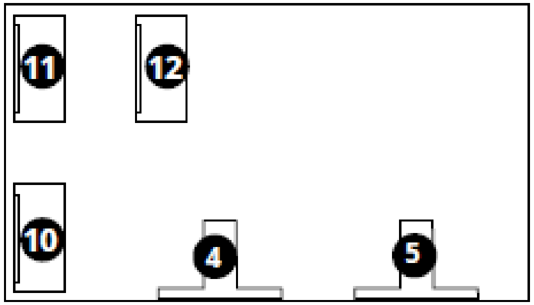
4. Rack module PCB

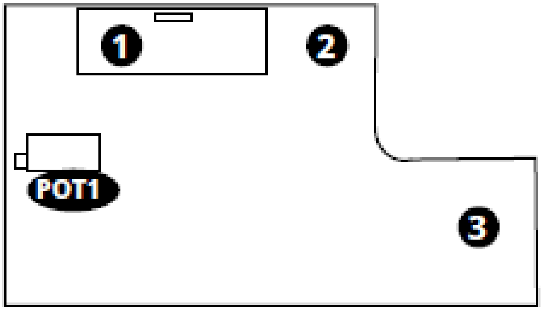


PCB layout

1 Rack puller motor connector
2 Rack aligner motor interface board connector
3 Level detector interface board connector
5 Rack mover motor connector
7 Internal bus cable
9 Barcode reader connector
11 Power connector from PCB7 (Master)
13 Rack position infra LED connector
14 Innermost catch position sensor connector
16 Outermost catch position sensor connector
23 Tube presence sensor connector
25 Rack mover full sensor connector
28 Rack before load sensor connector
30 Rack aligner sensors board to board connector
35 Programming bus cable
SW1 PCB address DIP switch (only slider 3 is in the down position)
SW2 Configuration DIP switch (with only slider 10 on)

4. Rack module PCB 2





11 Rack aligner motor board-to-board connector
12 Rack aligner motor connector
10 Rack aligner sensors board-to-board connector
4 Aligner close sensor
5 Aligner open sensor

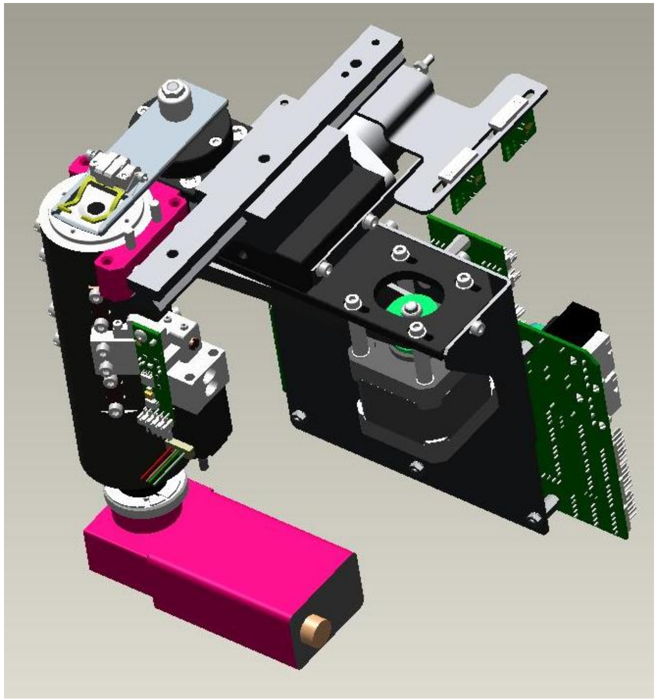
1 16-pin board-to-board connector
2 Level detector antenna pin connector
3 Rack position sensor (other side of PCB)
POT1 Potentiometer controlling the antenna

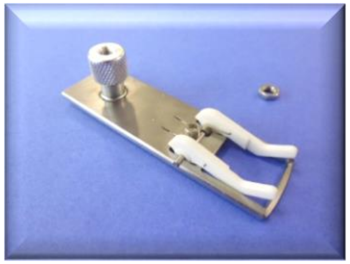
5. Microscope module

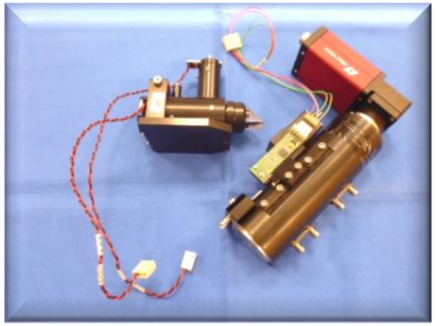
Main components

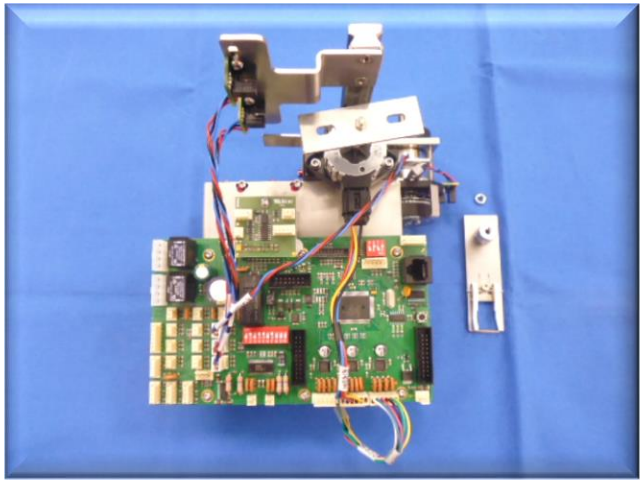
- microscope arm
- microscope lamp
- microscope and camera

(objective, controller motor, CCD camera)









Module description:

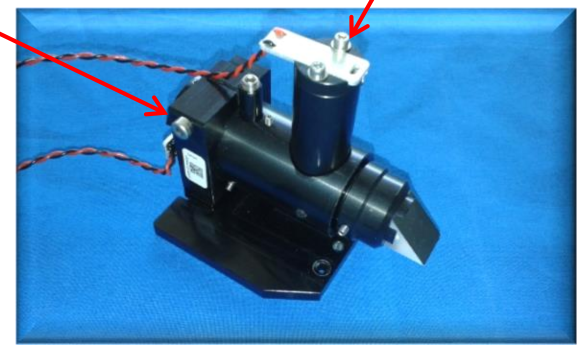
The microscope module is responsible for taking the images of the sample. By default it takes 15 images.

5. Microscope unit and lamp

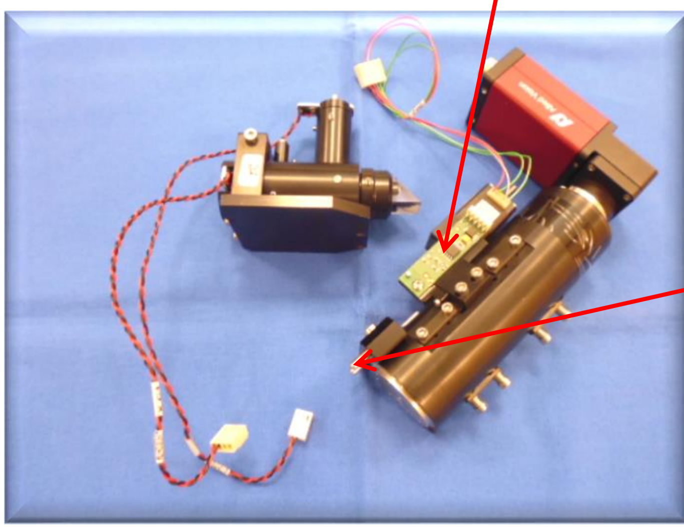
Phase contrast microscope and lamp

Phase Contrast LED

Brightfield LED



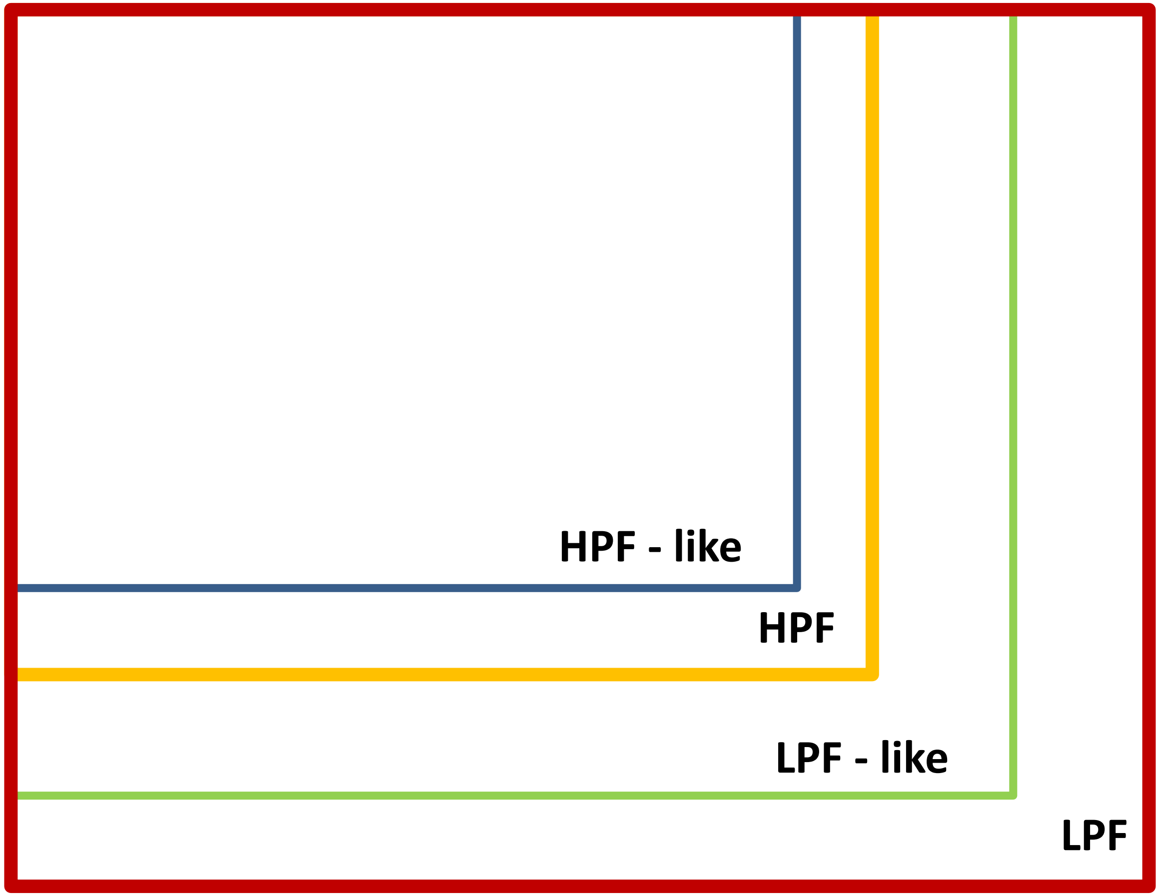
Interface PCB



Lamp positioning pins

The microscope module and the lamp is not field serviceable: the FSE should return the full module set if a problem occurs with them.

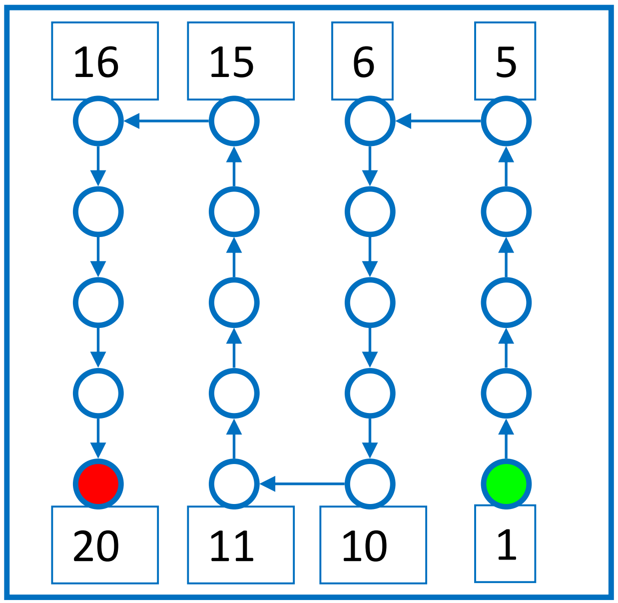
5. Viewfield

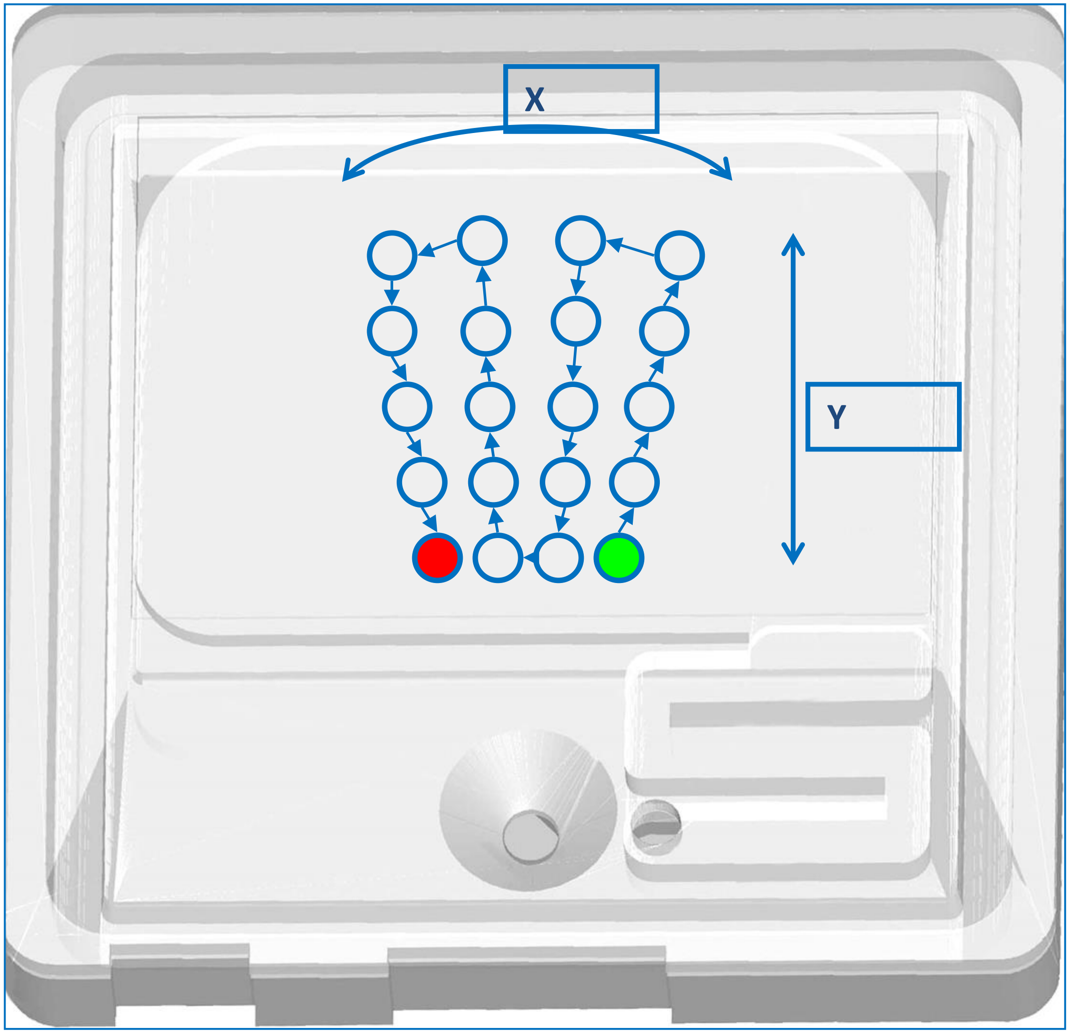


HPF like X 1.5 = HPF
LPF like X 0.5 = HPF

HPF = High Power Field
LPF = Low Power Field

5. Image positions





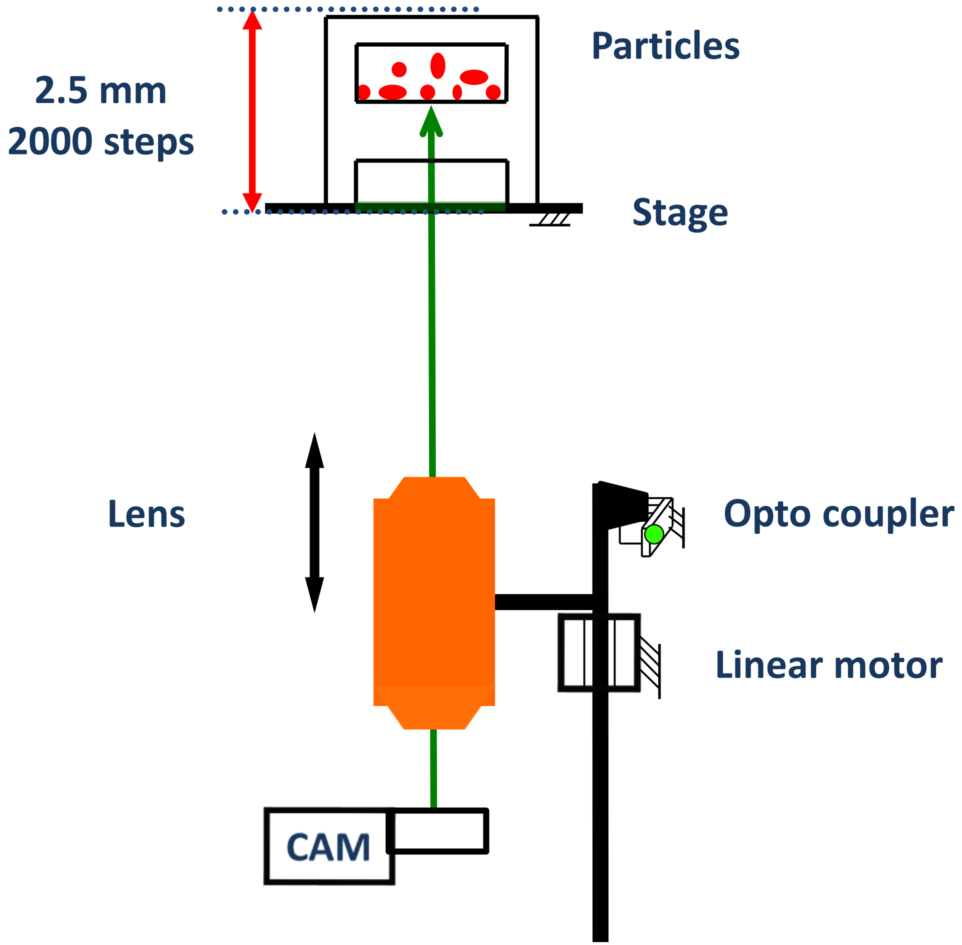
5. UriSed 2FL/Mini, UriSed 3 and UriSed 3 PRO microscope parameters

	Urised 2 FL / Mini	Urised 3	Urised 3 PRO
Full focus range:	2.5 mm / 2000 steps	2.5 mm / 2000 steps	2.5 mm / 2000 steps
Hard Focus			
Scanned range:	250 µm / 200 steps	250 µm / ~200 steps	250 µm / 200 steps
Stepping by:	2	3	2
No. of images:	100	67	100
Hard focus start:	Z-100 steps	Z-100 steps	Z-100 steps
Fine Focus			
Scanned range:	40 µm / 32 steps	40 µm / 32 steps	40 µm / 32 steps
Stepping by:	2	3	2
No. of images:	16	11	16

See notes for further details about the values

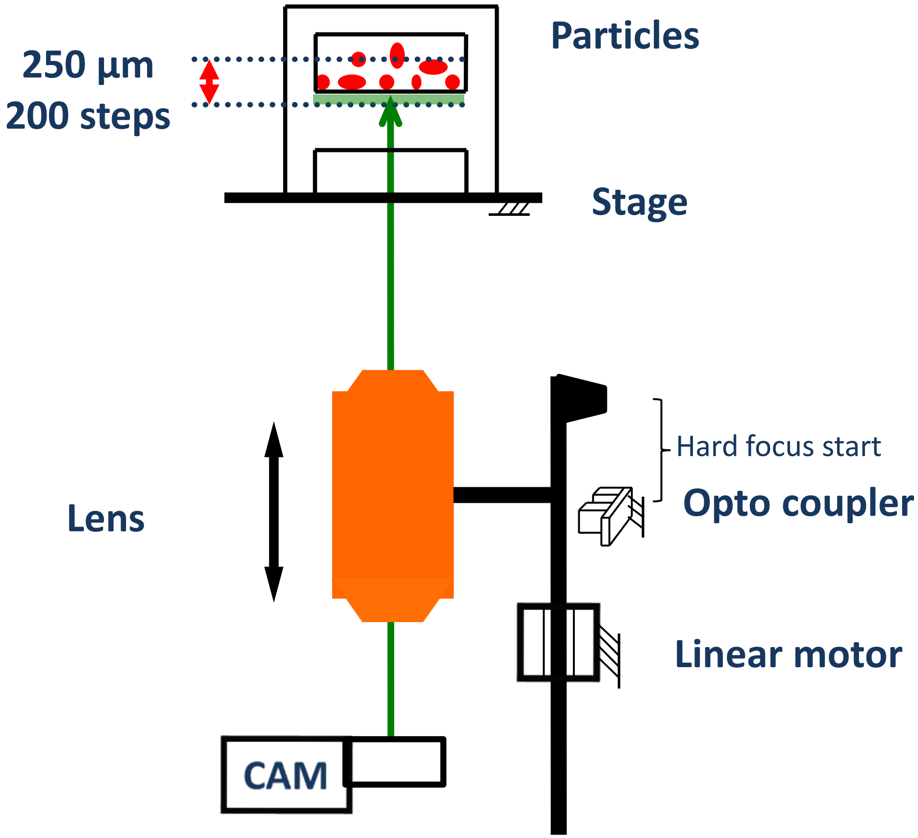
5. Microscope module

Focusing - Full scan

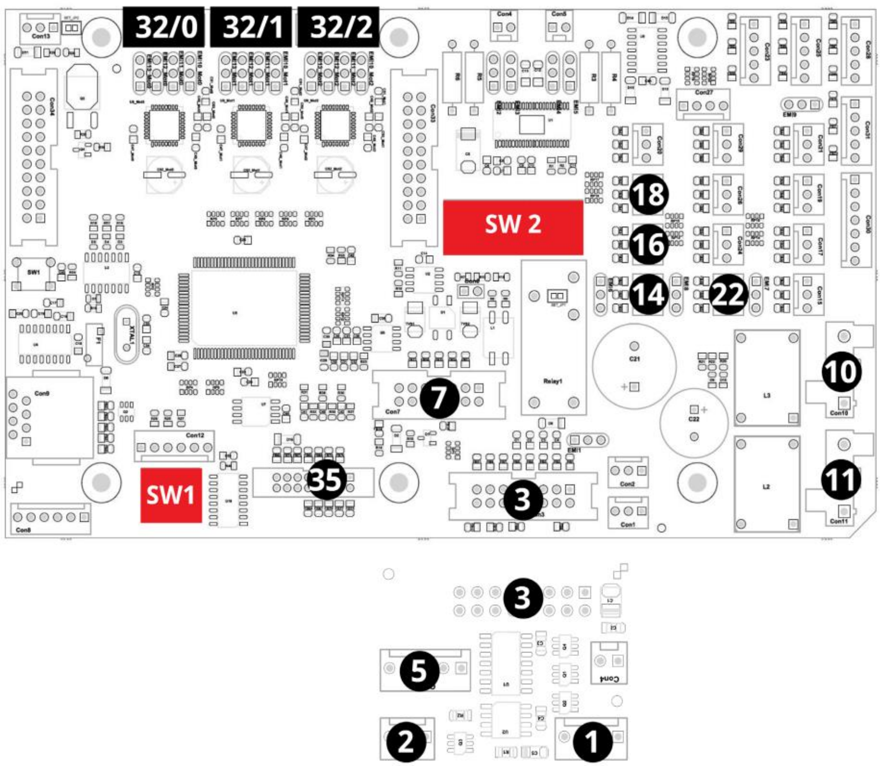


5. Microscope module

Focusing – Hard focus



5. Microscope module PCB



PCB layout

1 Phase-contrtast light connector
2 Brightfield light connector
3 Extension circuit board connector
5 Focus motor home position sensor connector
7 Internal bus cable connector
10 Power connector cable running to PCB 7 (master)
11 Power connector cable coming from the power supply
14 Microscope arm rotary home position sensor
16 Microscope arm linear home position sensor connector
18 Microscope arm linear end position sensor connector
22 Waste bin micro switch sensor connector
32/0 Microscope arm rotary motor connector
32/1 Microscope arm linear motor connector
32/2 Focus motor connector
35 Programming bus cable
SW1 PCB address DIP switch (sliders 1 and 3 are in the down position)
SW2 Configuration DIP switch (with sliders 2, 4, 5, and 7 on; all other sliders are off)

6. Centrifuge module

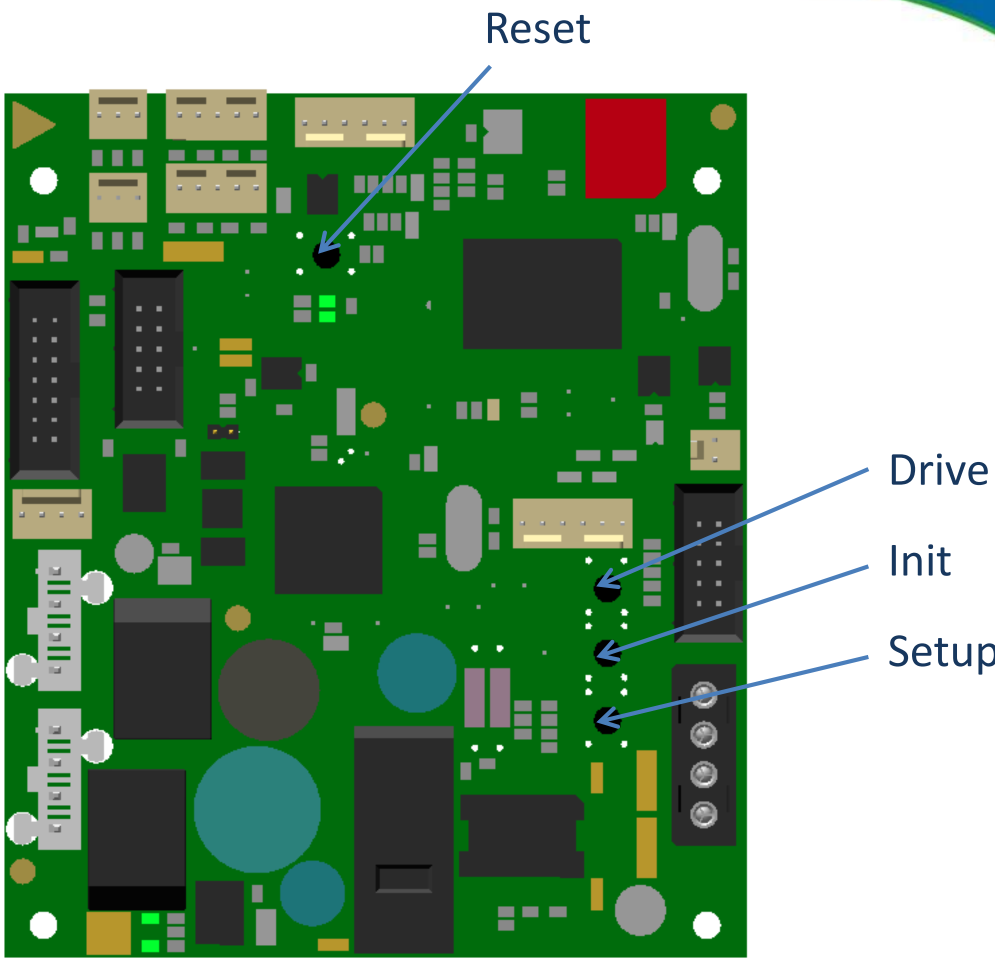
Main components
•DC motor
•Encoder
•Centrifuge arm

Functions
- Monolayering the sample
- 10 seconds, 2000 RPM

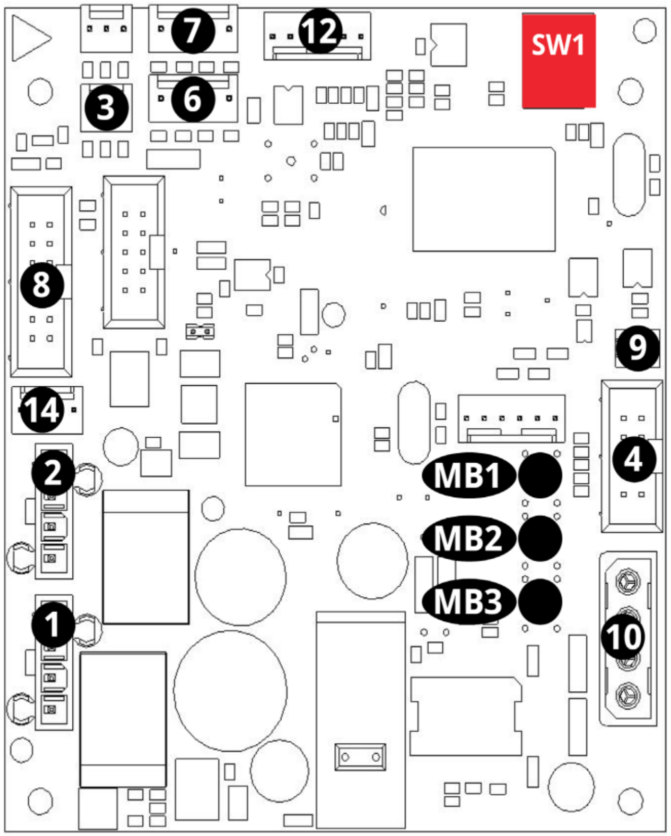
Centrifuge control card
•The relay powers-off the motor when the centrifuge house is removed
• In direct connection to the power supply by a cable

Micro switch door sensors
Two micro switches, one is for the software signal and one works as a hardware interlock

Transparent centrifuge house
• New centrifuge cover to meet international standards



6. Centrifuge module PCB



PCB layout

1 Power connector coming from the power supply
2 Power connector going towards the robot module PCB
3 'Power on' standby power cable connector
4 Rotary encoder sensor signal connector
6 Back cuvette rail inductive sensor connector
7 Centrifuge cover detector
8 Internal Bus cable connector
9 NC
10 Centrifuge motor connector
12 Centrifuge programming cable connector to Master PCB
14 Centrifuge interlock
MB1 'Drive' motor cycle mini pushbutton
MB2 Initializing mini pushbutton
MB3 Null (base) position setup mini pushbutton
SW1 PCB address DIP switch (sliders 2 and 3 are in the down position)

7. Master PCB

Attached components

• Door sensor

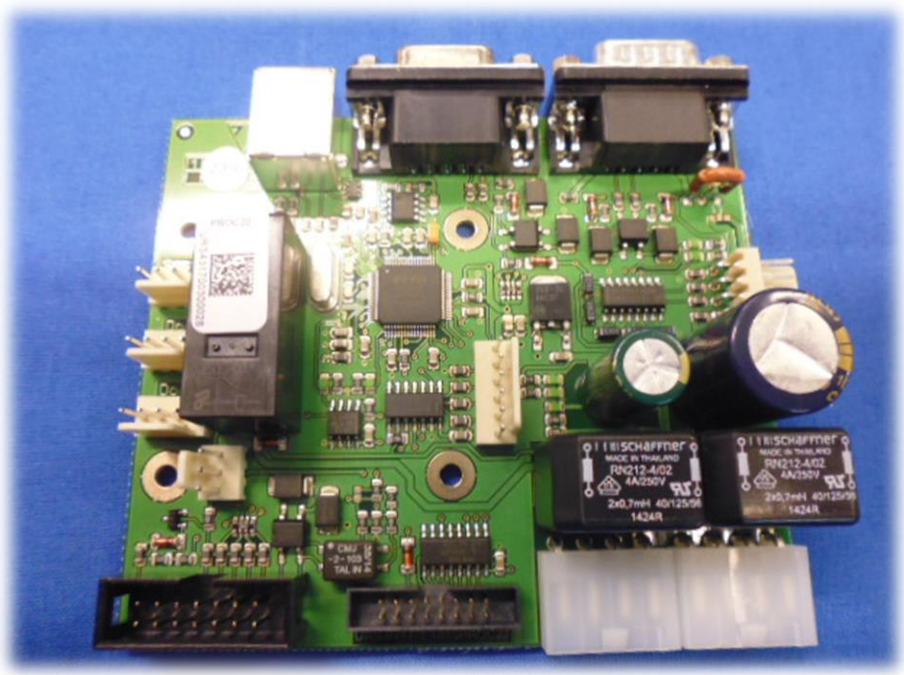
• Interlock switch

Functions

- Two-way Interface between PC and PCBs

-Sensor connectors of liquid containers

-Type B USB connector to Device



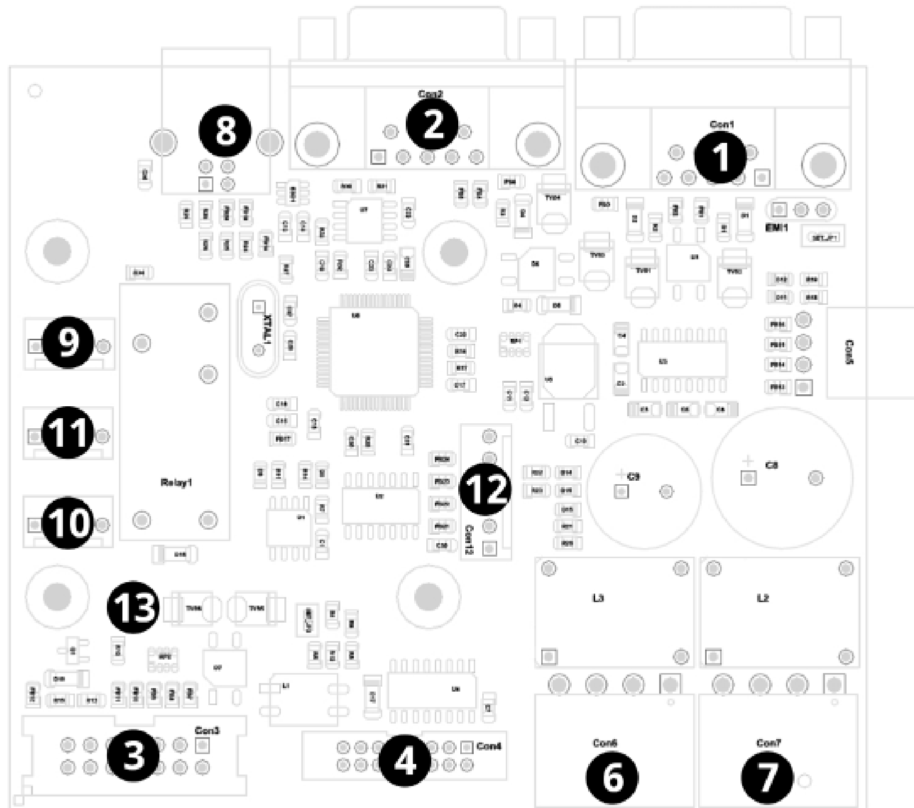
Module description:

The Master module is responsible for the communication between the UriSed 3 PRO instrument and the PC. (The installation of the FTDI driver is needed for the recognition of the card by the PC. After that the configuration of the FTDI driver is necessary for the proper microscope operation.)

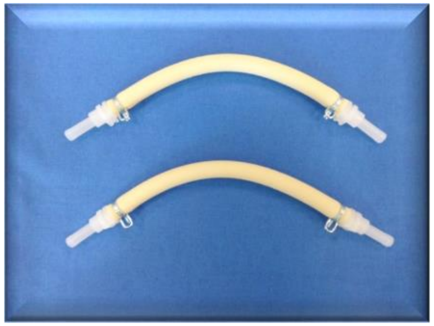
7. Master PCB

PCB layout

1 Waste container sensor connector
2 IF water container sensor connector
3 Bus cable connector
4 Programming bus cable
6 Power cable connector coming from PCB 5 (microscope)
7 Power cable connector to PCB 4 (rack)
8 USB Type B connector to PC
9 Service switch connector
10 NC
11 Door safety relay connector
12 Programming cable connector to PCB 6 (centrifuge)
13 ‘Power on’ standby power cable connector



Service maintenance parts



Neoprene tubing for peristaltic pumps

Filter

Preventive Maintenance:
Once a year for systems running up to 150 samples/day
Twice a year for systems running up to 400 samples/day

Pipes

1. PIPE_SUC1 +PU_ATM6/ATM4X240MM
2. PIPE_SUC2 +PU_ATM6/ATM4X330MM
3. PIPE_GRA1 +PU_ATM6/ATM4X150MM
4. PIPE_GRA1 +PU_ATM8/ATM6X240MM
5. PIPE_GRA2 +PU_ATM8/ATM6X80MM
6. PIPE_DIR1 +PU_ATM6/ATM4X220MM
7. PIPE_DIR1 +PU_ATM6/ATM4X430MM
8. PIPE_DIR2 PU_ATM6/ATM4X210MM(200MM)
9. PIPE_DIR2 +PU_ATM6/ATM4X210MM
10. PIPE_DIR3 +PU_ATM8/ATM6X70MM
11. PIPE_OUT3 +PU_ATM6/ATM4X180MM
12. PIPE_OUT4 +PU_ATM6/ATM4X140MM

Common troubleshooting

Cuvette Empty (Results marked with black square):

- Adjust pipetting position of the probe
- Check for Bubbles in the tubing system
- Load a new cuvette cartridge from an other LOT and double-check

No cuvette present:

- Clean the detector window under the Front Guide
- Cuvette value may always be above 850 (the cut value)-> fine-tune it by adjusting the potentiometer
- Replace the detector

Blurry images:

- Clean the microscope lens window
- Confirm cuvette is completely pushed into the microscope arm and adjust stage position setup if needed
- Load a new cuvette cartridge from an other LOT and double-check

Instrument skips tubes:

- Check the Rack handler level to confirm rack is moving smoothly (not crocked) in and out of the analyzer
- Check the tube position (in-out) correction and adjust it if necessary
- Adjust/Replace the tube presence sensor

Common troubleshooting 2.

Instrument turns off within few seconds after power on:

- Confirm power sequence (Computer turned before Main unit)
- Confirm Master and centrifuge PCB connectors are tightly connected on the boards (connectors may be faulty and require replacement)
- Master or Centrifuge PCB failure (sw failure)
- Replace the Power Supply

Connection Error:

- Confirm Main unit is turned on
- Confirm USB cable is connected from PC to Main Unit
- Replace Power Supply

PCB (1, 2, 3, 4, 5, 6, 7) Communication Errors:

- Check the USB cable between the PC and the instrument and if it has ferrite ring attached to it. Replace if necessary.
- Confirm BUS connector is tightly connected to PCB
- Check if the power connector is tightly connected to PCB
- Confirm dipswitches are on/off correctly for PCB
- Replace Power Supply (5V failure)
- Replace Master PCB